

CAOSMO

Polyphonic chaos synthesizer

USER MANUAL

Version 0.5 · November 2026

Eloy Platas

Free software under the GNU GPL v3

Contents

1 **Introduction** 4
What it offers 4
Origin 4
What's new in version 0.5 4
What's new in version 0.4
Requirements 5

2 **Installation** 6
macOS and Gatekeeper 6
Windows 6
User files 6
Building from source 6

3 **Chaos for musicians** 8
Dynamical systems 8
Attractors 8
Sensitivity to initial conditions 8
Bifurcations 8
Synchronisation and locking 8
How Caosmo protects the system 8

4 **Architecture and signal flow** 9
Voices and voice stealing 9
Per-note seed 9

5 **The interface** 10
Top bar 10
Bottom bar 10
The four pages 11
Working with controls 11
Container menus 11
Undo 11
Settings 12

6 **The lanes** 13
LANE 13
DRIFT 14
PITCH 14
SUBS 14

7 **The twelve original engines** 15
Kuramoto 15
Duffing 16
Sprott 17
Lorenz + Rössler 17
Logistic / Hénon 18
Chua 18
Gray-Scott 19
Lorenz Additive 19
Wave Grains 20
Bowed String 20
Mackey-Glass 21
Circle Map 22

8 **The SuperCollider engines** 23
Gendy 23
Maps 24
Ecosystems 25
Noise 26
Membrane 26

9 **Chance: triggers, distributions and new waves** 28

Random triggers ×4 28
Distributions 29
LFO waves with memory 29

10 New filters, attractors and effects 31
Friction and Spring filters 31
New attractors 31
Spectral and cycle effects 31

11 Filter, envelope, EQ and output 33
FILTER 33
ENV GEN 34
Lane EQ 35
OUTPUT 35

12 Modulation 37
LFO ×8 37
VENV ×4 39
Attractors ×4 40
Matrix ×24 41
Keytrack 42
Macros 43
MIDI, MIDI Learn and MPE 43

13 Master 44

14 Effect buses 46

15 DELUGE FX 48
Mod FX 48
EQ and destruction 48
Delay 49
Reverb 49
Compressor 49
Stutter and volume 50

16 Presets 51
The browser 51
Saving 52
The 20 categories 52
How they were made 53

17 Tutorials 54
A pad that synchronises 54
A double-well bass with bite 54
Sand-and-metal percussion 54
A cloud that rewrites itself 54
An expressive MPE cello 55
A techno drone with the Deluge 55
Writing a bifurcation 55
Tuned rain with Poisson triggers 55

18 Performance and CPU 56

19 Troubleshooting 57

20 How it was made 58
Research 58
Code architecture 58
The Deluge port 58
Presets 58
Testing 59

21 Glossary 60

22 Appendices 61
Preset catalogue 61
Parameter names in the DAW 62

Version history 62
Licence 63
Credits 63

1

Introduction

Caosmo is a polyphonic synthesizer that plays no fixed waveforms and no samples. Each note starts a **dynamical system**, and what you hear is that system's trajectory, solved sample by sample. Oscillators that synchronise with each other, non-linear springs pushed to their limit, strange attractors, iterated maps, a chaotic electronic circuit, a chemical reaction on a ring of cells, a bowed string, a delay loop with memory and a map that meshes frequencies like the teeth of a gear.

The result is an instrument that behaves more like a physical object than an oscillator. It has calm regions and turbulent regions, it responds differently depending on how you arrive at a setting, and it never sounds exactly the same twice.

1.1 What it offers

Block	Contents
Voices	Up to 64 voices. Each note plays through **3 independent lanes**.
Engines	**17 chaotic and stochastic systems**, plus *None* to switch a lane off.
Per lane	2 sub-oscillators, analogue drift, a 14-type filter, an envelope with 4 modes (ADSR, Vactrol, LFO, VENV), a 5-band EQ with 4 snapshots and morph, drive with 5 flavours, 3 sends.
Modulation	8 linkable LFOs, 4 drawable envelopes (VENV), 4 attractors of 12 types, 4 random trigger generators, 10 distributions (one drawable), a 24-route matrix, a drawable keytrack, 5 macros, MIDI learn and MPE.
Master	EQ with analyser, saturator, compressor with sidechain, width and a true-peak limiter.
Effects	3 buses with 3 slots (12 effect types, 4 of them spectral) and the **DELUGE FX** rack with the effects from the open-source Synthstrom Audible Deluge firmware.
Presets	288 factory presets in 20 categories, with tags, favourites and search.
Formats	VST3 and a standalone application for Linux, Windows and macOS (universal).

1.2 Origin

Caosmo is a functional reimplementation of the concept described in the public documentation of **Fragua** by Pura Belia: a three-lane-per-note chaos synthesizer with twelve engines. All the code, names, interface, presets and this manual are original; it contains no code or assets from Fragua. It is an independent work with no connection to Pura Belia.

1.3 What's new in version 0.5

Everything new comes from studying the SuperCollider documentation for chaos and randomness generators:

- Five new engines: **Gendy** (Xenakis), **Maps** (ten chaotic maps), **Ecosystems** (neurons, chemical clocks, predators and prey), **Noise** and **Membrane** (a ball bouncing on a modal body).
- Logistic/Henon can iterate in time with the note and play in tune.
- Four random **trigger generators** (Dust, Gauss, Coin, Lorenz, Clock) with direct actions.

- Ten probability **distributions**, one of them drawable, for LFOs, triggers and Note Random.
- LFO waves with memory: Brownian, Geo Brownian, Noise Ring and Markov.
- **Friction** and **Spring** filters, seven new attractor types and six effects: Diffuse, Mag Noise, Rand Comb, Noise Synth, Wave Loss and Squiz.
- 32 new presets in three categories: Stochastic, Neurons & Maps and Bouncing Bodies.

Nothing breaks: 0.4 presets and projects sound the same.

1.4 What's new in version 0.4

- **DELUGE FX** rack with Mod FX, EQ, destruction, delay, three reverbs, compressor and stutter from the Deluge firmware.
- Caosmo is now distributed under the **GNU GPL v3**.
- **Windows and macOS** binaries as well as Linux, built and validated automatically.
- Constant 1.5 ms latency (the DAW no longer sees jumps when the limiter is switched).
- Fix: the Deluge Mod FX could crash the plugin when Depth was raised from 0.

1.5 Requirements

- A **VST3** host (REAPER, Bitwig, Ableton Live, Cubase, Studio One, Ardour...) or the standalone version. Logic Pro does not load VST3: on a Mac, use another DAW or the standalone app.
- Linux x86-64 (glibc 2.35 or later, e.g. Ubuntu 22.04), 64-bit Windows 10/11, or macOS 10.15 or later (Intel or Apple Silicon).
- A modern CPU: every lane solves differential equations per sample. Read the performance chapter before asking for 64 voices on three lanes.

2

Installation

Each package contains the `Caosmo.vst3` plugin, the standalone version, the licence and the manual. The 288 presets are built into the plugin: there is nothing else to copy.

System	Copy `Caosmo.vst3` to	Standalone
Linux	`~/.vst3/` (your user only) or `/usr/lib/vst3/` (system-wide)	`Caosmo-standalone`
Windows	`C:\Program Files\Common Files\VST3\`	`Caosmo.exe`
macOS	`~/Library/Audio/Plug-Ins/VST3/` or `/Library/Audio/Plug-Ins/VST3/`	`Caosmo.app`

Then rescan plugins in your DAW. In REAPER: *Options → Preferences → Plug-ins → VST → Re-scan*. Caosmo appears as an instrument by the manufacturer **Eloy Platas**.

2.1 macOS and Gatekeeper

The binary is universal and carries an *ad hoc* signature, not an Apple developer certificate. If macOS says the plugin "is damaged" or "can't be opened", remove the quarantine flag from Terminal:

```
xattr -dr com.apple.quarantine ~/Library/Audio/Plug-Ins/VST3/Caosmo.vst3
xattr -dr com.apple.quarantine /Applications/Caosmo.app
```

2.2 Windows

Copying to `Common Files\VST3` needs administrator rights. If your DAW does not find the plugin, check that this folder is in its list of VST3 paths. The standalone version opens *Options → Audio/MIDI Settings* to choose a device; ASIO gives the lowest latency.

2.3 User files

What	Where
User presets (`.caosmo`)	`Documents/Caosmo Presets/User`
User VENV shapes	`Documents/Caosmo Presets/VENV`
Favourites and interface preferences	Application settings (per user)

"Documents" is `~/Documents` on Linux and macOS and the Documents folder of your profile on Windows. Presets are XML text files: you can share them and keep them under version control.

2.4 Building from source

Caosmo is free software (GPL v3). To build it you need CMake 3.22+, a C++23 compiler (GCC 13, Clang 16, AppleClang 15 or Visual Studio 2022's clang-cl) and JUCE 8.0.4:

```
git clone https://github.com/eloyplatas/CAOSMO-VST-linux-osx-windows.git caosmo
git clone --depth 1 --branch 8.0.4 https://github.com/juce-framework/JUCE.git JUCE
cd caosmo
cmake -B build -DCMAKE_BUILD_TYPE=Release -DCAOSMO_TESTS=OFF
cmake --build build --config Release
```

On Windows add `-A x64 -T ClangCL` to the first `cmake` call: the Deluge code uses GCC/Clang extensions that MSVC does not accept. On Linux you need the ALSA, X11, Freetype and fontconfig development headers. The repository builds all three platforms on GitHub Actions on every change and leaves the packages on the `builds` branch.

3

Chaos for musicians

You don't need maths to play Caosmo, but five ideas will save you a lot of time.

3.1 Dynamical systems

A dynamical system is a set of variables (position, velocity, temperature, phase...) and a rule saying how they change over time. A sine oscillator is the most boring dynamical system possible: it goes round in a circle forever. Caosmo's engines are systems with **non-linearities**: cubic terms, absolute values, saturation, products between variables. Those non-linearities are what make complex behaviour possible.

3.2 Attractors

Let a system with friction run, and its state ends up in a particular region: the **attractor**. It can be a point (the system stops: silence), a cycle (the system repeats a shape: a periodic tone), a torus (two superimposed cycles: a tone with beating) or a **strange attractor** (the system never repeats yet never escapes: a living, rough or noisy sound). The SCOPE's phase portraits literally show each lane's attractor.

3.3 Sensitivity to initial conditions

In a chaotic system, two almost identical states drift apart exponentially. That's why Caosmo starts each note with a **different seed**: two identical notes in a row are not identical, and a chord does not sound like transposed copies. If you want repeatability, several engines have a **Note → Free** menu that keeps the state between notes instead of resetting it.

3.4 Bifurcations

As you move a control parameter (K, Gamma, Chaos, Alpha, Sharp, R...) the kind of attractor can change suddenly: a tone splits into its sub-octave (*period doubling*), a beating appears, or the sound breaks into noise. Those points are **bifurcations**. They are the instrument's best material: modulate slowly around them with a VENV or an LFO and you will hear the system make up its mind. Many boundaries have **hysteresis**: the point where the system breaks on the way up is not where it recovers on the way down.

3.5 Synchronisation and locking

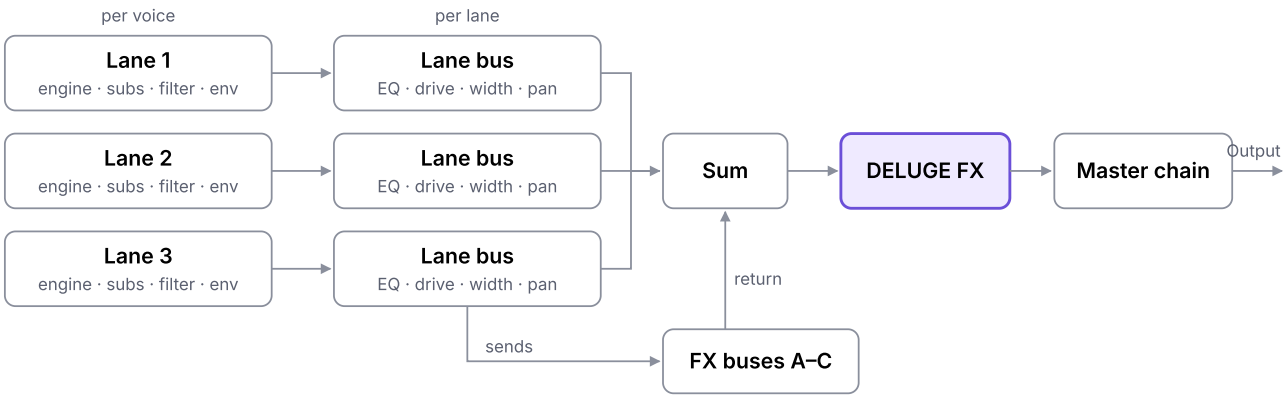
Several engines (Kuramoto, Circle Map, Mackey-Glass) try to **lock**: oscillators that end up beating together, frequencies that get trapped in simple ratios (1:1, 2:3, 3:4). Between the locking zones there are quasi-periodic and chaotic regions. That's where those engines get their metallic, vocal or trembling character.

3.6 How Caosmo protects the system

A non-linear system can run off to infinity. Every engine watches its state: if a variable blows up or stops being a valid number, the engine quietly restarts from a safe position. At worst you hear a small click, never a burst of level. All pitched engines also limit their integration step so they stay stable on high notes.

4

Architecture and signal flow



Every note you play takes a **voice**. Inside the voice, each lane that has an engine has its own complete chain:

1. **Engine**: produces the signal by solving its system, tuned to the note if it is a pitched engine.
2. **Sub 1** (another instance of the same engine, 1–3 octaves down) and **Sub 2** (a sine 1–2 octaves down) are added to the engine.
3. **Filter** with 14 types.
4. **Envelope** (VCA, and in LPG mode the filter too), scaled by velocity.

The voices of each lane are then summed on the **lane bus**, where **EQ**, **drive**, **width**, **bass mono**, **pan**, **level** and the **sends** to FX buses A–C are applied. The lane buses and the FX returns are summed and pass through the **DELUGE FX** rack (if enabled) and the **master chain**.

This split matters for modulation: whatever happens inside the voice (pitch, filter, engine, envelope, level) can be modulated **per note**; whatever happens on the lane bus (EQ, drive, width, pan, sends) can only be modulated by **global sources**, because separate notes no longer exist there.

4.1 Voices and voice stealing

Voices sets the maximum polyphony (1, 4, 8, 16, 32 or 64). When they run out, Caosmo first takes a free voice, then the oldest released voice, and as a last resort the oldest sounding voice. With **Voices = 1** the synth is monophonic: with **Legato** on, tied notes do not retrigger the envelopes, and **Glide** slides the pitch (*Fingered*: only on tied notes; *Always*: always).

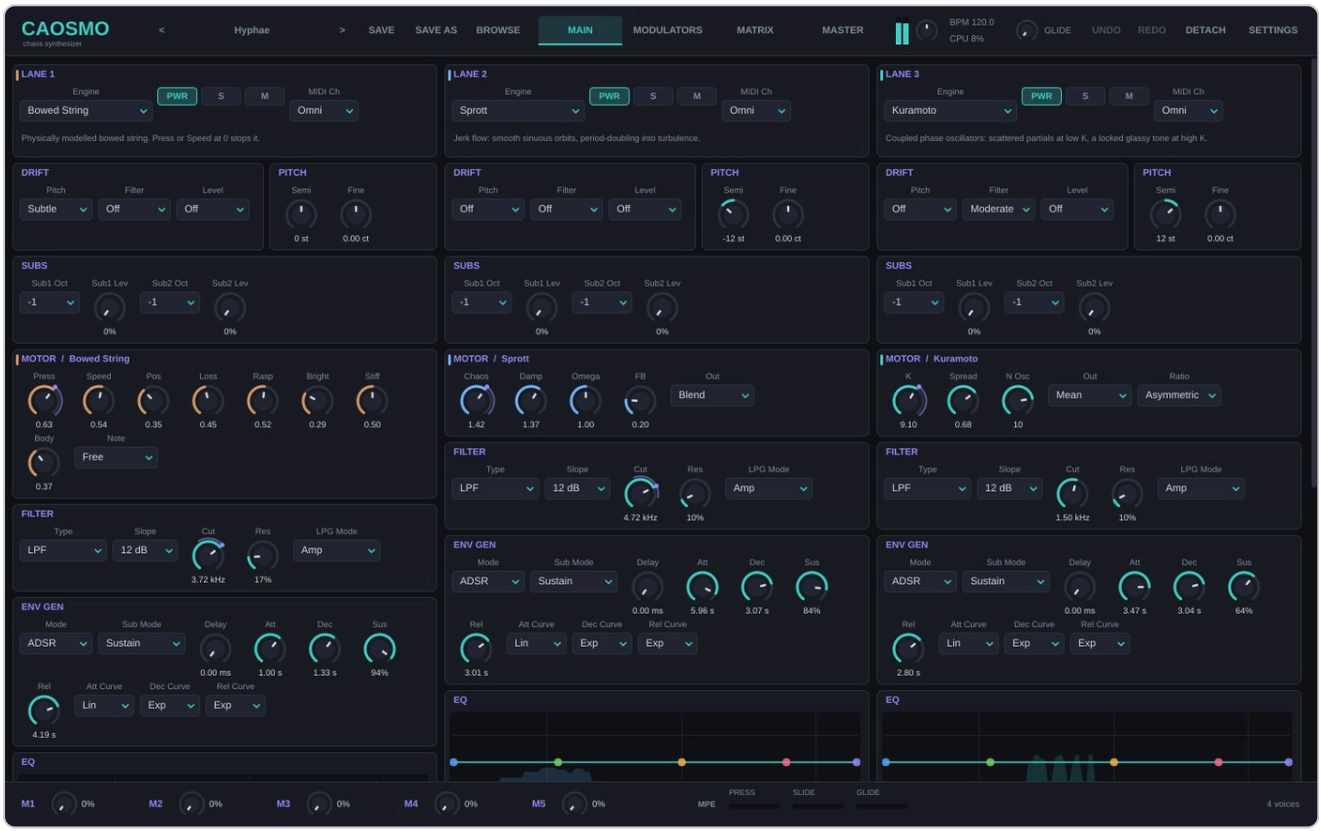
4.2 Per-note seed

Each note gets its own random number, used by the engines for their initial state, by drift, by the *Note Random* source and by the stereo placement of Voice Spread. That's why a preset feels alive even without modulation.

5

The interface

The window is 1440 × 900 points at 100 % and can be scaled from 75 to 200 % in *Settings*. Long pages scroll with the wheel or the side bar.

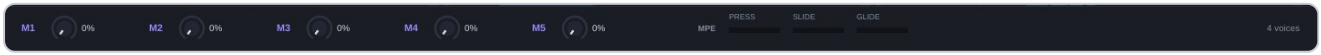


5.1 Top bar



- **< preset name >**: click the name to open the browser; the arrows load the previous and next preset in the filtered list.
- **SAVE / SAVE AS / BROWSE**: save over the current user preset, save as a new one, and open the browser.
- **MAIN, MODULATORS, MATRIX, MASTER**: the four pages.
- **Meter, output level, BPM and CPU**: output level, Output Gain, host tempo and audio engine load.
- **GLIDE**: portamento time.
- **UNDO / REDO**.
- **DETACH**: opens the current page in a floating window, handy with two screens.
- **SETTINGS**: settings, presets and credits.

5.2 Bottom bar



The **macros M1–M5**, the **MPE** meters of the latest note (PRESS, SLIDE, GLIDE) and the number of sounding voices.

5.3 The four pages

Page	Contents
MAIN	The three lanes in columns: LANE, DRIFT, PITCH, SUBS, MOTOR, FILTER, ENV GEN, EQ, SCOPE and OUTPUT.
MODULATORS	8 LFOs, 4 VENVs with their editor and 4 attractors with their 3-D portrait.
MATRIX	The 24 routes, the KEYTRACK editor and the MACROS view.
MASTER	Tabs CHAIN (master chain and meters), FX (buses A–C) and DELUGE (the Deluge rack).

5.4 Working with controls

- **Drag** to change a value; **Shift + drag** for fine adjustment.
- **Double-click** opens numeric entry.
- **Mouse wheel**: adjusts according to *Settings* (Fine, Coarse or Off, so scrolling never moves anything).
- **Right-click**: the control's menu with *Modulate with…* (pick a source and create the route), *Send to Macro*, *Connected* (lists and deletes the routes that reach the control), *MIDI / MPE → MIDI Learn*, default value and numeric entry.
- **Modulation halo**: the violet arc around a control shows the range its routes cover; the bright dot is the modulated value right now (of the latest note for per-note destinations).

5.5 Container menus

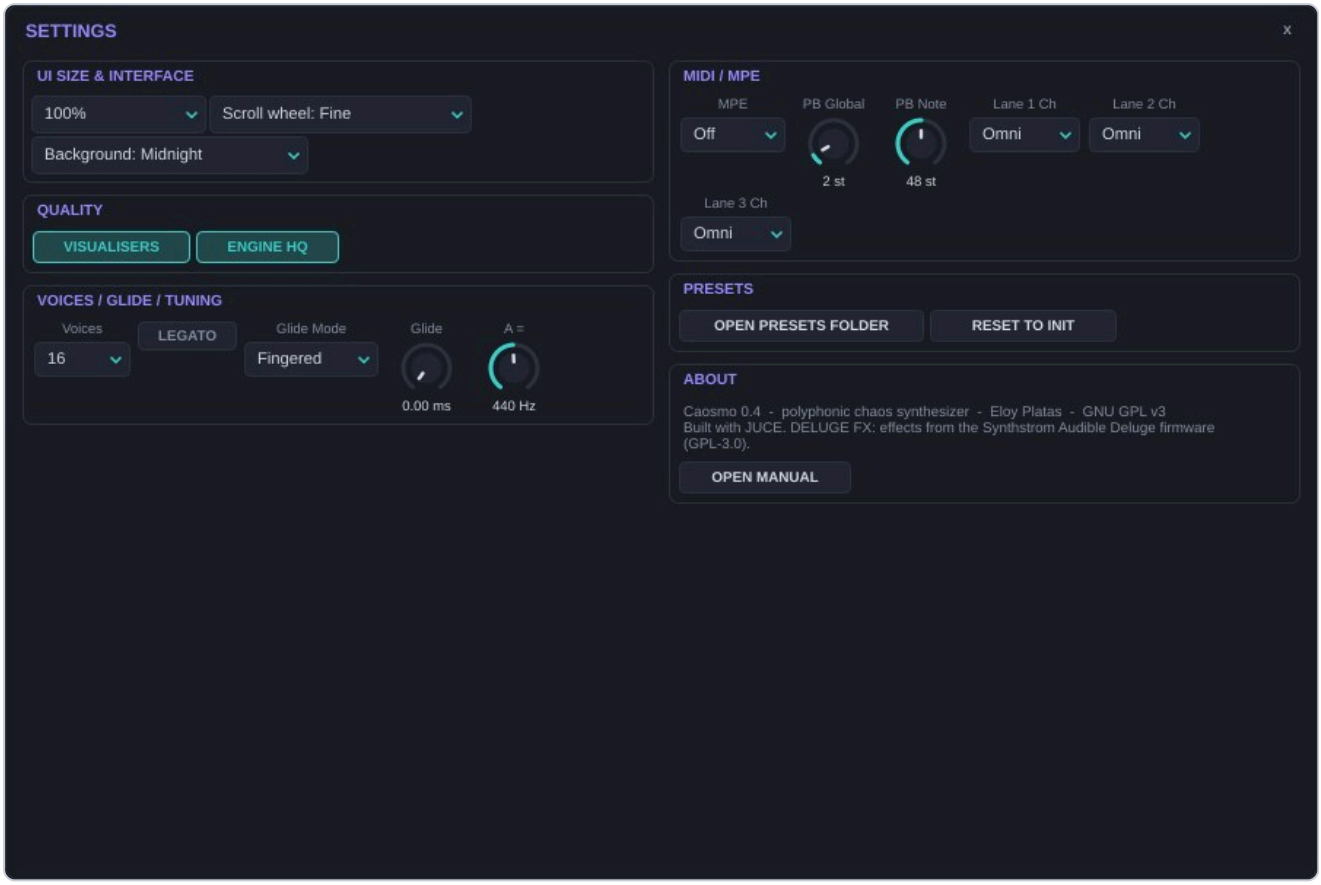
The title of each block (FILTER, ENV GEN, LFO 3…) is a menu:

- **Copy / Paste**: copies the block's settings; *Paste with routes* includes the modulation routes that reach it.
- **Vary**: moves each value randomly by up to ±15 %. Perfect for variations on a sound that already works.
- **Randomise**: completely random values within useful ranges.
- **Copy to Lane** (LANE only): copies the whole lane to another lane.

5.6 Undo

Ctrl + Z / Ctrl + Shift + Z (Cmd on Mac) undo and redo changes to parameters, routes, VENV shapes and the keytrack.

5.7 Settings

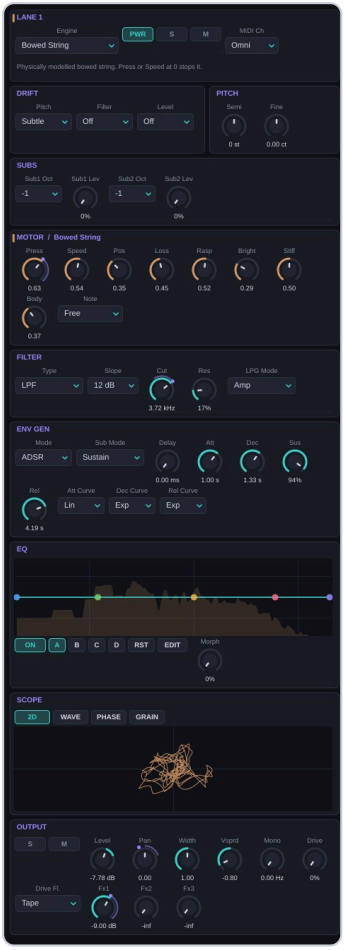


Section	Setting	What it does
UI Size & Interface	Size	75–200 % interface scale.
	Scroll wheel	Fine, Coarse or Off (so scrolling never moves controls).
	Background	Midnight or Graphite.
Quality	VISUALISERS	Runs or pauses scopes and visualisers (saves UI CPU).
	ENGINE HQ	×2 oversampling in the engines that use it (Lorenz Additive).
Voices / Glide / Tuning	Voices, Legato, Glide Mode, Glide, A =	Polyphony, legato, portamento and reference tuning (415–466 Hz).
MIDI / MPE	MPE, PB Global, PB Note	MPE mode and global and per-note pitch-bend ranges.
	Lane 1–3 Ch	MIDI channel of each lane (same setting as in LANE).
Presets	OPEN PRESETS FOLDER	Opens the preset folder in your file manager.
	RESET TO INIT	Replaces the patch with the initial one.
About	OPEN MANUAL	Version, licence and this manual.

Interface preferences (size, wheel, background, visualisers) belong to you and are not stored in presets.

6

The lanes



Caosmo has three identical lanes (L1, L2, L3). Each note plays through every lane that has an engine and is powered. Use them as three stacked synths, as one sound and its sub-layers, or split them by MIDI channel for a keyboard split or a simple multitimbral setup.

6.1 LANE

Parameter	Range (default)	What it does
Engine	None + 17 engines (L1 Kuramoto, L2 and L3 None)	The lane's engine. The settings of all 17 engines are kept: you can switch engines and come back without losing anything.
PWR	On/Off (On)	Switches the lane off: it stops being computed and its envelope stops modulating.
S / M	Solo / Mute	Isolate or silence the lane while it keeps running (its sources keep modulating).
MIDI Ch	Omni, 1–16 (Omni)	Channel that triggers the lane. Ignored when MPE is on.

6.2 DRIFT

Every note gets a fixed random deviation when it starts, like the tolerances of an analogue synth. It does not change during the note.

Level	Pitch	Filter	Level
Off	0	0	0
Subtle	±3 cents	±1 semitone	±0.5 dB
Moderate	±8 cents	±3 semitones	±1.5 dB
Heavy	±20 cents	±7 semitones	±3 dB

Subtle on all three lanes thickens a chord without detuning it; *Heavy* on pitch gives the detuned chorus of an old tape.

6.3 PITCH

Pitch transposes ±36 semitones and **Fine** ±100 cents. Both are per-note modulation destinations: a slow LFO on Fine gives vibrato; a VENV on Pitch, pitch sweeps. On unpitched engines (Lorenz+Rossler, Logistic/Henon, Gray-Scott without BPF) Pitch only affects the sub-oscillators and any filters that follow the note.

6.4 SUBS

Parameter	Range (default)	What it does
Sub1 Oct	−1, −2, −3 oct (−1)	Octave of Sub 1.
Sub1 Lev	0–100 % (0)	**A second instance of the same engine**, with the same settings, playing 1–3 octaves down. Adds weight without losing character. Doubles the engine's CPU cost.
Sub2 Oct	−1, −2 oct (−1)	Octave of Sub 2.
Sub2 Lev	0–100 % (0)	A **pure sine** with a stereo detune of about ±3 cents. Gives a fundamental to noisy or unpitched engines. Almost free.

Both subs go through the same filter and envelope as the engine.

7

The twelve original engines

Each engine solves a different system. **Pitched engines** scale their time base to the MIDI note; **unpitched engines** run at their own speed and give textures, noise and internal rhythms. To tune the latter, use Sub 2, the Comb or Formant filter, or Gray-Scott's built-in BPF.

#	Engine	Pitched	Character	CPU
1	Kuramoto	Yes	Swarms, choirs, glass that locks	Medium
2	Duffing	Yes	Basses, flutes, leaps between wells	Low
3	Sprott	Yes	Sinuous leads, period doubling	Low
4	Lorenz+Rossler	No	Growls, wind, organic textures	Low
5	Logistic/Henon	No	Granular noise, dust, clicks	Very low
6	Chua	Yes	Raspy buzzes, aggressive basses	Low
7	Gray-Scott	No (yes with BPF)	Crackle, cells, rain	Medium
8	Lorenz Additive	Yes	Breathing spectral pads	High
9	Wave Grains	Yes	Grain clouds, living wavetables	Medium–high
10	Bowed String	Yes	Bowed strings, bells, piano wire	Medium
11	Mackey-Glass	Yes	Feedback tones, crackle	Low
12	Circle Map	Yes	Gears, intervals that lock	Low

Engines 13–17 (Gendy, Maps, Ecosystems, Noise and Membrane), added in version 0.5, have their own chapter: *The SuperCollider engines*.

The first six parameters of each engine are the matrix destinations *Motor 1–6*, in the order of the tables below.

7.1 Kuramoto

The system. In 1975 Yoshiki Kuramoto described how a population of oscillators with different frequencies can synchronise when coupled: fireflies that end up flashing together, pacemaker cells in the heart, applause that falls into step. Each oscillator turns at its natural frequency ω_i and is pulled towards the group's mean phase:

$$d\theta_i/dt = \omega_i + K \cdot r \cdot \sin(\psi - \theta_i)$$

where r (0–1) measures how closely the phases agree and ψ is the mean phase.

In Caosmo. Up to 12 phase oscillators. Their natural frequencies are spread around the note according to **Ratio** and **Spread**, and normalised so that their mean is the note. With low K each oscillator goes its own way: you hear a cluster of beating partials. As K rises there is a critical threshold where the group locks and the sound becomes a single, clean, glassy tone. That threshold has hysteresis: raising and lowering K does not pass through the same sounds.

Parameter	Range (default)	What it does
K	0–15 (2)	Coupling strength. It scales with the note and with Spread, so the threshold is similar across the keyboard.
Spread	0–1 (0.5)	Spread of the natural frequencies. More spread: partials further apart, and a higher K needed to lock.
N Osc	3–12 (6)	Active oscillators. Those entering or leaving fade over 20 ms, so it can be modulated without clicks.
Out	Mean, Coherent (Mean)	*Mean*: the sum of all. *Coherent*: the sum weighted by how synchronised they are; almost silent when scattered, full when locked.
Ratio	Asymmetric, Linear, Golden, Harmonic, Random (Asymmetric)	Frequency layout. *Linear*: evenly spaced. *Asymmetric*: quadratic, denser at the bottom. *Golden*: golden-ratio positions, maximally inharmonic. *Harmonic*: 1, 1+Spread, 1+2·Spread…; with Spread = 1 it is the harmonic series. *Random*: fixed but irregular.

Sweet spots. Harmonic with Spread 1 and K between 2 and 6: an organ that detunes and pulls itself back together. Golden with K near the threshold and Out = Coherent: bells that appear and vanish. Modulate K with a slow LFO or an attractor for pads that drift in and out of phase (preset *Glass Synchrony*).

7.2 Duffing

The system. A spring whose stiffness grows with displacement and which has two rest positions (a **double well**), pushed by a periodic force:

$$\ddot{x} + \delta\dot{x} - x + k \cdot x^3 = \gamma \cdot f(\omega t)$$

With little force the mass oscillates inside one well: a round tone. With more force it leaps from one well to the other, sometimes regularly (sub-harmonics) and sometimes chaotically.

In Caosmo. The time base follows the note; the force oscillates at **Omega** times the note. Integration is fourth-order Runge-Kutta per sample.

Parameter	Range (default)	What it does
Gamma	0–1 (0.3)	Force amplitude. The main chaos control: from a steady tone to leaps.
Delta	0.01–1 (0.1)	Friction. Low: resonant and wild. High: damped and tame.
Omega	0.25–4 (1)	Force frequency relative to the note. 1 = resonance; 0.5 and 2 give sub-harmonics and harmonics; non-integer values give beating.
Stiff	0.5–3 (1)	Non-linear stiffness. Moves the wells and the brightness.
FB	0–1 (0)	Adds the position to the force: self-excitation, less stable.
Out	Position, Velocity, Blend (Position)	The variable you hear. *Velocity* is brighter (it is the derivative).
Drive	Sine, Square, Triangle, Saw, Noise (Sine)	Shape of the force. *Square* and *Saw* excite more harmonics; *Noise* (filtered at Omega × note) gives a breathy, flute- or wind-like sound.

Sweet spots. Bass: Gamma 0.2–0.35, Delta 0.15, Omega 1, Out Position (preset *Double Well Bass*). Flute: Drive Noise, Gamma 0.5, Out Velocity (*Duffing Flute*). Modulate Gamma with velocity so that playing hard makes the mass jump between wells.

7.3 Sprott

The system. Julien C. Sprott searched for the simplest possible chaotic systems. This one is a *jerk* flow (the derivative of acceleration):

$x''' = -a\cdot\ddot{x} - \dot{x} + |x| - 1$

With a high, the system traces a clean cycle; as a falls it goes through a cascade of **period doublings** (the tone drops to the sub-octave, then the sub-octave of the sub-octave...) until it reaches chaos.

In Caosmo. Chaos, Damp and FB combine into the coefficient a, always inside the window where the system does not run away. The time base follows the note and **Omega** multiplies it.

Parameter	Range (default)	What it does
Chaos	0.1–3 (1)	Lowers the coefficient a: from periodic tone to doublings and turbulence.
Damp	0.1–3 (1)	Raises a: more damped, more stable. The counterweight to Chaos.
Omega	0.25–4 (1)	Multiplies the time base: 2 = roughly an octave up (the period depends on the attractor).
FB	0–1 (0)	Adds a sinusoidal term that twists the orbit and limits the maximum chaos.
Out	X, Y, Z, Blend (X)	The coordinate you hear. X is the smoothest; Z the roughest.

Sweet spots. Raise Chaos slowly from 0.1 with Damp at 1: you will hear every period doubling. A long VENV on Chaos is the heart of the *Bifurcations* category.

7.4 Lorenz + Rössler

The system. Two of the most famous strange attractors. Lorenz (1963), a simplified model of atmospheric convection with its two "butterfly wings":

$\dot{x} = \sigma(y - x)\ \dot{y} = x(\rho - z) - y\ \dot{z} = xy - \beta z$ ($\rho = 28$, $\beta = 8/3$)

Rössler (1976), a spiral that folds back on itself:

$\dot{x} = -y - z\ \dot{y} = x + a\cdot y\ \dot{z} = b + z(x - c)$ ($a = b = 0.2$)

In Caosmo. Both run at fixed speeds (Lorenz slow and growling, Rössler around 190 cycles per second) and are mixed. **They do not follow the note**: use Sub 2, a resonant filter or Pitch on the subs to give them pitch.

Parameter	Range (default)	What it does
Lorenz Lvl	0–1 (0.7)	Level of the Lorenz attractor.
Rossler Lvl	0–1 (0.7)	Level of the Rössler attractor.
Sigma	5–20 (10)	Lorenz σ: how fast x follows y. Higher: more abrupt jumps between wings.
C	4–8 (5.7)	Rössler c. 4: a period-2 cycle, almost a tone. 5.7: classic chaos. 8: spiky spirals, brighter.
Lorenz Out	X, Y, Z, Blend (X)	Lorenz coordinate. X changes sign when it switches wings: the characteristic "growl".
Rossler Out	X, Y, Z, Blend (X)	Rössler coordinate. Z is very impulsive.

Sweet spots. Rössler only, C 4.2 and a resonant LPF: an almost-tuned hum that breaks up as C rises. Lorenz Out Z through an HPF gives gusts of wind.

7.5 Logistic / Hénon

The system. Two **maps**: rules that compute the next value from the current one. The logistic map (May, 1976) is the most famous population equation in chaos theory:

$x_{n+1} = r \cdot x_n \cdot (1 - x_n)$

The Hénon map (1976) is its two-dimensional cousin:

$x_{n+1} = 1 - a \cdot x_n^2 + y_n \; y_{n+1} = b \cdot x_n$

In Caosmo. With *Clock* on *Per Sample* (the default) both maps are iterated **once per sample**. At 48 kHz that produces structured noise: grainy, dusty, with fixed high tones whenever the map falls into a periodic window (r ≈ 3.83, for example, gives a period-3 cycle). In that mode it does not follow the note; with the *Note* clocks it does (see the table).

Parameter	Range (default)	What it does
Morph	0–1 (0.5)	Mix of the logistic map (0) and the Hénon map (1).
R	3.6–3.95 (3.9)	Logistic parameter. Below 3.57 there is no chaos; between 3.6 and 3.95 there is chaos with periodic windows.
A	0.8–1.4 (1.4)	Hénon parameter. 1.4 is the classic attractor; lower values give periodic orbits: high tones.
B	0–0.4 (0.3)	Second Hénon parameter: how much the map remembers the previous value.
Smooth	0–1 (0)	Low-pass from 20 kHz down to 200 Hz with level compensation: from hiss to rumble.
Drift	0–1 (0)	Slow oscillation (about 6 s) of R and A: the noise passes through periodic windows and back.
Iter Oct	−2 – 7 (3)	With a *Note* clock: iterations per note cycle, in octaves.
Clock	Per Sample, Note Step, Note Linear, Note Cubic (Per Sample)	*Per Sample*: one iteration per sample (noise). The *Note* clocks iterate at the note × 2^Iter Oct with stepped, linear or cubic output: the map plays in tune.

Sweet spots. Smooth 0.7 and a resonant BPF tuned to the note: sand percussion (*Sand Snare*). High Drift and Morph halfway: a radio between stations (*Radio Static*).

7.6 Chua

The system. Leon Chua's circuit (1983) was the first electronic circuit built to demonstrate chaos. It has a piecewise-linear non-linear resistor and produces the famous **double-scroll** attractor:

$\dot{x} = \alpha(y - x - f(x)) \; \dot{y} = x - y + z \; \dot{z} = -\beta \cdot y$

In Caosmo. Integrated per sample with Runge-Kutta 4. The time base follows the note; with **Omega** = 3.5 the fundamental lands on the note played.

Parameter	Range (default)	What it does
Alpha	8.5–11 (10)	Main control. Low: a cycle on a single scroll (a tone with harmonics). Medium: a chaotic single scroll. High: double scroll, the sound jumps between two centres.
Beta	13.5–18.5 (15)	Second parameter. Moves where Alpha's bifurcations fall.

Parameter	Range (default)	What it does
Omega	0.5–8 (3.5)	Time base. 3.5 = the note; 7 ≈ an octave up.

Sweet spots. Alpha 9 and a ladder LPF: a raspy, stable bass. Alpha sweeping 9.5–10.5 with a slow LFO: the sound moves from scroll to double scroll and back (*Scroll Sub*).

7.7 Gray-Scott

The system. A **reaction-diffusion** model: two chemicals (U and V) that diffuse and react (U + 2V → 3V). Depending on the feed rate F and the kill rate k you get spots that divide like cells, stripes, waves or extinction. It is the same model that explains the spots on some animals' skin.

In Caosmo. A ring of 24 cells updated once per sample. The sound is a spatial projection of chemical V and of its change: it crackles when the "cells" divide or die. It does not follow the note, except through its built-in BPF.

Parameter	Range (default)	What it does
Feed	0.02–0.06 (0.0367)	Feed rate F. Higher: more activity, patterns that fill in.
Kill	0.04–0.075 (0.0649)	Kill rate k. Higher: spots die out and the sound fades. F and k together choose the kind of pattern.
Drift	0–1 (0.3)	Slowly oscillates F and k and, if the pattern dies, seeds new spots. At 0 the sound can die out for good on that note.
BPF	0–1 (0)	Mixes in a band-pass tuned to the note: gives it pitch.
Q	0.5–15 (5)	Resonance of that band-pass.

Sweet spots. The defaults (F 0.0367, k 0.0649) sit in the mitosis zone: spots that divide. BPF 0.8 with Q 10: tuned drops (*Rain on Tin*, *Reaction Bells*).

7.8 Lorenz Additive

The system. Additive synthesis: 32 sine partials on the note's harmonic series. Their amplitudes are not set by an envelope but by a **slow Lorenz attractor**: each partial reads a different combination of the three coordinates, so the spectrum rearranges itself continuously without repeating.

In Caosmo. The partials have a natural roll-off (about $1/n^{0.85}$), and any that would exceed the Nyquist frequency are muted. With **Engine HQ** it is computed at twice the sample rate with a linear-phase decimation filter.

Parameter	Range (default)	What it does
Chaos	0.5–2 (1)	Attractor speed: from slow evolution to a boiling spectrum.
Density	4–32 (32)	Active partials. Fractional values fade the last one. Fewer partials: darker and hollower.
Smooth	0–1 (0.3)	Amplitude transition time, from 2 ms to about 600 ms. Low: shimmer. High: breathing.
Drive	0.1–2 (0.7)	Soft saturation of the sum. Low: clean. High: compressed and dense.

Sweet spots. Chaos 0.6, Smooth 0.8, Density 24: a choral pad (*Coherent Choir*, *Spectral Strings*). It is the most expensive engine: with many voices, switch Engine HQ off.

7.9 Wave Grains

The system. A two-dimensional wavetable (sine, triangle, saw and square in the four corners) played through a **cloud of grains**. An internal attractor (Lorenz, Rössler, Chua or Duffing) moves the table position and the fold, so the cloud rewrites itself.

In Caosmo. Up to 16 simultaneous grains tuned to the note, with anti-aliased (polyBLEP) oscillators.

Parameter	Range (default)	What it does
X / Y	0–1 (0.3 / 0.2)	Position in the table. (0,0) sine, (1,0) triangle, (0,1) saw, (1,1) square; mixtures in between.
PWM	0.05–0.95 (0.5)	Pulse width of the square.
Fold	0–1 (0)	Sine wavefolding: adds metallic harmonics.
Size ms	1–1000 ms (200)	Length of each grain.
Formant	0.25–16 (1)	Read speed inside the cycle, restarted every period of the note: shifts the formant without changing pitch (like hard sync). 1 = neutral.
Chaos Rate	0.02–20 Hz (0.4)	Speed of the internal attractor.
Chaos>X / >Y / >Fold	−1 to +1 (0.3 / 0.2 / 0)	How much the attractor moves each coordinate.
Tau	0–1 (0.4)	Smoothing of the attractor, from 5 ms to 0.8 s. Low: jumps; high: drift.
Count	1–16 (6)	Grains per grain length: the density of the cloud.
Grain Shape	0–1 (0.5)	Window: from bell (Hann, soft) to trapezoid (more presence, more click).
Scatter	0–1 (0.3)	Irregularity of the birth time and of the initial phase.
Jitter	0–1 (0.1)	Random detune per grain, up to ±0.5 semitones.
Spray	0–1 (0.4)	Random stereo placement of the grains.
Grain FB	0–1 (0)	Feeds the output back into the grains' phase: chaotic phase modulation.
Chaos	Lorenz, Rossler, Chua, Duffing (Lorenz)	Internal attractor.
Grain	Frozen, Continuous (Frozen)	*Frozen*: each grain keeps the position it was born with (layered texture). *Continuous*: all grains follow the current position (moving wavetable).
Reverse	Off/On (Off)	Plays the waveform backwards.

Sweet spots. Size 40 ms, Count 12, Scatter 0.8, Spray 1: a particle cloud (*Particle Fields*). Size 400 ms, Count 2, Continuous: a slow breathing wavetable. Switch the SCOPE to **Grain** to watch the grains.

7.10 Bowed String

The system. A physical model of a bowed string (digital waveguide): the string is a delay line reflected at the bridge and the nut, and the bow interacts with it through a non-linear *stick-slip* friction curve. That mechanism is chaotic by nature: too little pressure and the string whistles, too much and it scratches.

In Caosmo. Two slightly detuned strings (±2 cents, left and right) and a body with three modes (230, 600 and 1600 Hz).

Parameter	Range (default)	What it does
Press	0–1 (0.5)	Bow pressure. Below ~0.05 there is no contact (silence). Low: harmonics and whistles. High: a full tone, then scratching.
Speed	0–1 (0.5)	Bow speed (smoothed over 8 ms). 0 stops the string. Modulate it with velocity or aftertouch.
Pos	0–1 (0.3)	Contact point, from 4 % to 50 % of the string. Near the bridge (0): bright, *sul ponticello*. Towards the middle: sweet, *sul tasto*.
Loss	0–1 (0.4)	Loss per round trip. More: shorter, duller notes.
Rasp	0–1 (0.2)	Noise in the friction, proportional to speed: hair and rosin.
Bright	0–1 (0.5)	Low-pass inside the loop. More: more sustained highs.
Stiff	0–1 (0)	Dispersion (stiffness): the harmonics stretch upwards, towards piano wire, metal and bell.
Body	0–1 (0.4)	Body resonance mix.
Note	Retrig, Free (Retrig)	*Retrig*: the string starts at rest on each note. *Free*: it keeps vibrating (natural legato).

Sweet spots. Press 0.55, Speed 0.6, Pos 0.15, Body 0.6 with a Vactrol envelope: cello (*Cello Gesture*). Stiff 0.7, Loss 0.2, Speed on a short envelope: a struck string (*Piano Wire*, *Bell Foundry*). With MPE, route Pressure to Press and Slide to Pos.

7.11 Mackey-Glass

The system. In 1977 Michael Mackey and Leon Glass proposed a model of blood-cell production with a **delay**: what the body produces today depends on what was there a time τ ago.

$$dy/dt = \beta \cdot y(t-\tau) / (1 + y(t-\tau)^n) - \gamma \cdot y$$

With short delays the system oscillates; with high exponents n the oscillation doubles and becomes chaotic.

In Caosmo. The delay τ **tunes itself**: a period detector adjusts the loop until the oscillation matches the note. The result is feedback tones with stable pitch that can break into chaos.

Parameter	Range (default)	What it does
Drive	0–1 (0.5)	Production gain β. More: more harmonics and more instability.
Speed	0–1 (0.5)	Integration speed: changes the waveform (from rounded to sharp). The detector corrects the tuning.
Tap	0–1 (0.3)	Position of a second delay tap, from 5 % to 100 % of the loop.
Loss	0–1 (0.8)	Ratio of removal γ to production β. Low: a strong, rich oscillation, more prone to chaos. High: more damped and stable.
Sharp	0–1 (0.2)	Exponent n (5.5–14.5 in Tuned, 5.5–23.5 in Classic). High: square shape and the onset of chaos.
Bright	0–1 (0.5)	Adds the signal's derivative and saturation: more presence.
Crackle	0–1 (0)	Random kicks to the system: embers, crackle.
Tap2	0–1 (0.5)	Mix of the second tap: a two-delay system, more complex.
Res	0–1 (0)	Extra loop gain: more resonant and aggressive.

Parameter	Range (default)	What it does
Curve	Tuned, Classic (Tuned)	*Tuned*: smooth, stable curve. *Classic*: the original equation, sharper and 1.6 times faster.
Octave	Fixed, Free (Fixed)	If the system doubles its period: *Fixed* keeps the tuning and you hear the sub-octave underneath; *Free* retunes to the new period (the note can jump an octave).
Note	Retrig, Free (Retrig)	Restart the loop on each note or keep its state.

Sweet spots. Sharp 0.6, Crackle 0.15: a bass with embers (*Ember Bass*). Sharp on a VENV from 0.2 to 0.9: a tone that splits and bursts (*Feigenbaum Gate*).

7.12 Circle Map

The system. The circle map is the simplest model of two oscillators trying to lock, like the teeth of two gears or a pushed swing:

$\theta_{n+1} = \theta_n + \Omega - (K/2\pi)\cdot\sin(2\pi\cdot\theta_n)$

For each strength K there are ranges of Ω (the **Arnold tongues**) where the frequency ratio is trapped on a simple fraction: 1/2, 2/3, 3/4... Between them the motion is quasi-periodic and, with K above 1, chaotic.

In Caosmo. The map is iterated at a rate tied to the note and its phase becomes a waveform. A second map (the **slave gear**) runs at twice the rate and is coupled to the first.

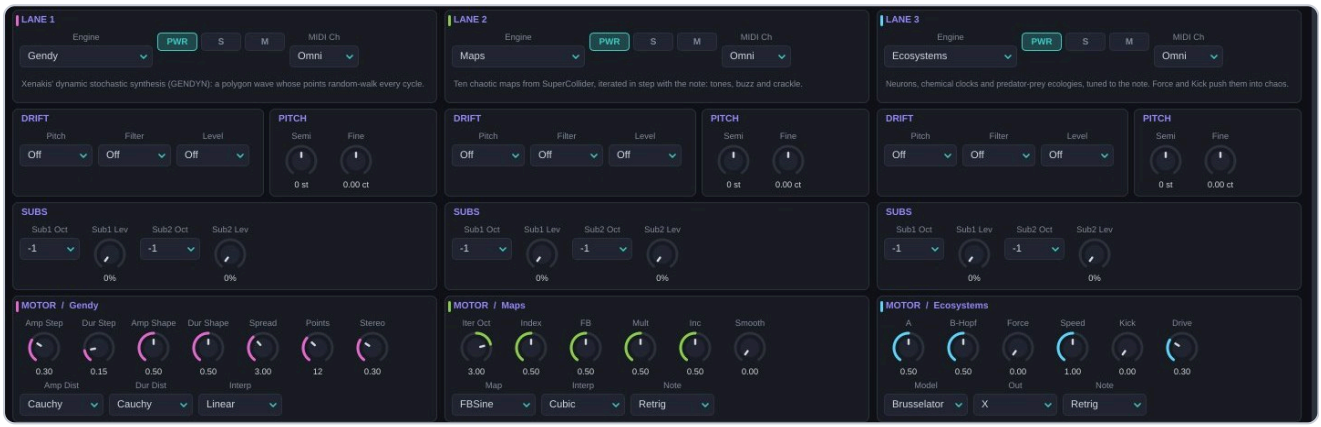
Parameter	Range (default)	What it does
Lock	0–1 (0.5)	Locking strength K. Low: free slipping. Around 0.9: the tongues cover almost everything and the tone locks. Higher: chaos.
Ratio	0–1 (0.5)	Frequency Ω. **0.5 = the note**. As you move it the pitch does not glide: it jumps from one fraction to the next, like a gearbox.
Bias	0–1 (0.5)	Shifts the phase of the locking term: changes which fractions grab most strongly.
Slew	0–1 (0.4)	Output low-pass, from 18 kHz down to 270 Hz.
Tongue	0–1 (0.2)	Widens the tongues and adds random instability: trembling locks.
Shape	0–1 (0.5)	Saturates the output wave: from sine to almost square.
Slave	0.5–2 (1)	Ratio of the slave gear to the main one.
Couple	0–1 (0.3)	Mutual coupling between the two maps and the slave mix (20 to 50 %).
Note	Retrig, Free (Retrig)	Restart the phases on each note or keep them.

Sweet spots. Lock 0.85, Ratio swept by a slow LFO: a melody of "locked" intervals (*Gear Train 16*, *Kalimba Gears*). Couple 0.7, Slave 1.5: two gears arguing (*Gear Shift Lead*).

8

The SuperCollider engines

Version 0.5 adds five engines (13–17) that come from studying the documentation of **SuperCollider**, the open-source synthesis language. They are reimplementations of its chaos and randomness generators (UGens), adapted to Caosmo's architecture: each one follows the note, gets its per-note seed and exposes its first six parameters as the matrix destinations *Motor 1–6*.



#	Engine	Pitched	Character	CPU
13	Gendy	Yes	Xenakis' stochastic synthesis: timbres that rewrite themselves every cycle	Low
14	Maps	Yes	Ten SuperCollider chaotic maps iterated in time with the note	Very low
15	Ecosystems	Yes	Neurons, chemical clocks and predator-prey populations	Low
16	Noise	No (yes with Key BPF)	Classic and chaotic noise: pink, brown, gray, Crackle, Dust	Very low
17	Membrane	Yes	A ball bouncing on a membrane, bar or bell	Medium

With 8 notes on three lanes, each of these engines uses 5 to 13 % of one core.

8.1 Gendy

The system. Iannis Xenakis' *dynamic stochastic synthesis* (GENDYN), in Nick Collins' Gendy1 version. The wave is a polygon of N points. Every cycle, each point takes a random step in amplitude and in duration, with a size drawn from a probability distribution. Barriers reflect the steps so the polygon cannot run away. The result has a recognisable pitch but a timbre that never repeats.

In Caosmo. The polygon's total duration is tuned to the note; *Spread* lets each segment stray a few semitones. Two independent generators feed the left and right channels.

Parameter	Range (default)	What it does
Amp Step	0–1 (0.3)	Size of the amplitude steps. Low: almost static. High: rough noise.
Dur Step	0–1 (0.15)	Size of the duration steps: how much the cycle wobbles and detunes.

Parameter	Range (default)	What it does
Amp Shape	0–1 (0.5)	Parameter of the amplitude distribution (the shape of the bell, of the tail...).
Dur Shape	0–1 (0.5)	The same for the duration distribution.
Spread	0–24 st (3)	Maximum deviation of each segment from the note.
Points	2–32 (12)	Polygon points. Few: harsh, bright tones. Many: finer timbres.
Stereo	0–1 (0.3)	Difference between the two generators: 0 mono, 1 two different voices.
Amp Dist / Dur Dist	Linear, Cauchy, Logist, Hyperbcos, Arcsine, Expon, Sinus (Cauchy)	Step distribution. Cauchy has long tails: small steps almost always and a huge one now and then. *Sinus* takes its steps from an internal sine.
Interp	Linear, Cubic (Linear)	How the points are joined. Cubic rounds the corners.

Sweet spots. Amp Step 0.1–0.2 with Dur Step 0.05: a breathing choir (*Xenakis Choir*). Points 6 with both steps above 0.5: the rough sound of *S709*. *Hyperbcos* is quieter than the other distributions: make up with Level.

8.2 Maps

The system. SuperCollider's *ChaosGen* family: ten iterated maps, each a rule that computes the next value from the current one.

In Caosmo. The map is iterated at the note frequency times 2^Iter Oct. Between iterations the output is held (Step), interpolated in a straight line (Linear) or along a Catmull-Rom curve (Cubic): SuperCollider's N, L and C suffixes. With a low Iter Oct you hear each iteration as part of the waveform; with a high Iter Oct the map makes a high timbre over the note. Parameters A–D change meaning with the map; the labels in the interface update themselves. With A–D at 50 % every map uses SuperCollider's default values.

Map	A	B	C	D	Character
Latoocarfian	a	b	c	d	From Pickover's book: tones, cycles or chaos depending on the four parameters.
Standard	K	—	—	—	Chirikov's map. K below 1: near-periodic hum. Above: it tears.
Gingerbread	X0	Y0	—	—	The Gingerbreadman: only the initial conditions matter; moving them restarts the map.
Cusp	a	b	—	—	Sharp peaks, a reedy timbre.
Quad	a	b	c	—	General quadratic map.
LinCong	Mult	Inc	Mod	—	Linear congruential generator: 8-bit digital staircases.
FBSine	Index	FB	Mult	Inc	A sine whose phase is advanced by a congruential map. Index 0.5, FB 0 and Mult 0.25 give a pure sine.
Crackle	Param	—	—	—	The Crackle map: crackling near collapse.
Logistic	R	—	—	—	The logistic map, pitched.
Henon	A	B	—	—	The Hénon map, pitched.

Parameter	Range (default)	What it does
Iter Oct	−2 – 7 (3)	Iterations per note cycle, in octaves: 3 = 8 iterations per cycle.

Parameter	Range (default)	What it does
A, B, C, D	0–1 (0.5)	Map parameters (see table).
Smooth	0–1 (0)	Low-pass with level compensation.
Map	The 10 maps (Latoocarfian)	System.
Interp	Step, Linear, Cubic (Cubic)	Interpolation between iterations.
Note	Retrig, Free (Retrig)	Retrig restarts the map on every note; Free lets it run on.

Logistic/Henon now follows the note too. Engine 5 has a new parameter, *Clock*. *Per Sample* is the old behaviour (one iteration per sample, unpitched noise). *Note Step*, *Note Linear* and *Note Cubic* iterate it like Maps, at the note times 2^*Iter Oct*: the grainy noise becomes a tuned tone.

8.3 Ecosystems

The system. Six models from biology and chemistry, taken from Nick Collins' SLUGens and Till Bovermann's ChaosUGens:

- **FitzHugh-Nagumo**: a simplified neuron. Below a current threshold it does nothing; above it, it fires regular spikes.
- **Brusselator**: a theoretical chemical clock. Past the Hopf bifurcation it oscillates on a limit cycle.
- **Oregonator**: the model of the Belousov-Zhabotinsky reaction, the famous chemical cocktail that changes colour in cycles.
- **Lotka-Volterra**: prey and predators. When prey are plentiful the predators grow, then eat them.
- **Arneodo-Coullet-Tresser**: three competing species; it has a chaotic attractor.
- **Spruce Budworm**: outbreaks of a forest pest, iterated as a map.

In Caosmo. The equations are integrated with 4th-order Runge-Kutta. A period detector measures the oscillation and adjusts the speed so it sounds at the note's pitch, even when P1 and P2 change the system's natural period. *Force* adds sinusoidal forcing at the note and *Kick* adds noise hits: both push the system off its cycle towards chaos.

Model	P1	P2
FitzHugh-Nagumo	Current: input current	Eps: recovery speed
Brusselator	A	B-Hopf: distance to the bifurcation
Oregonator	F: stoichiometry	Eps: time-scale separation
Lotka-Volterra	Prey: prey growth	Pred: predator death
Arneodo-Coullet-Tresser	Alpha	Immig: immigration
Spruce Budworm	Step: map step	K1: forest capacity

Parameter	Range (default)	What it does
P1, P2	0–1 (0.5)	Model parameters (see table). The labels change with the model.
Force	0–1 (0)	Periodic forcing at the note frequency.
Speed	0.25–4 (1)	Multiplies the speed: changes the timbre without changing the pitch it tunes to.
Kick	0–1 (0)	Noise impulses that knock the system off its orbit.
Drive	0–1 (0.3)	Output saturation.

Parameter	Range (default)	What it does
Model	The 6 models (FitzHugh-Nagumo)	System.
Out	X, Y, Z, Blend (X)	Variable you hear (Z only exists in ACT).
Note	Retrig, Free (Retrig)	Retrig restarts the system on every note; Free lets it run on.

Sweet spots. FitzHugh-Nagumo with P1 near the centre and a little Kick: a neuron that sometimes misfires (*Spiking Neuron*). At the extremes of P1 the neuron rests and the lane goes silent: raise Kick to wake it. Brusselator with Force 0.35: in and out of its cycle (*Brusselator Pad*).

8.4 Noise

The system. SuperCollider's noise generators: WhiteNoise, PinkNoise (Voss-McCartney algorithm), BrownNoise, GrayNoise (one bit of a register flipping at random), ClipNoise, Crackle and the Dust and Dust2 impulses.

Parameter	Range (default)	What it does
Tone	−1 – 1 (0)	Spectral tilt: dark to bright.
Density	1–20 000 /s (300)	Impulses per second for Dust and Dust2.
Chaos	1–2.05 (1.9)	Crackle's parameter. Near 2 it crackles; above, it breaks into bursts.
Stereo	0–1 (0.6)	Decorrelation between channels.
Key BPF	0–1 (0)	Mix of a band-pass tuned to the note: gives the noise a pitch.
Q	0.5–30 (6)	Band-pass resonance.
Type	White, Pink, Brown, Gray, Clip, Crackle, Dust, Dust2 (White)	Generator.
Clock	Free, Note Step, Note Smooth (Free)	*Note Step*: sample-and-hold at twice the note (tuned "digital" noise). *Note Smooth*: the same, interpolated.

Sweet spots. Dust2 with Key BPF 0.8 and Q 18: tuned rain (*Dust Devils*). Gray with Note Step and Q 25: a bit-flip lead (*Gray Code*).

8.5 Membrane

The system. A falling, bouncing ball, like Dan Stowell's TBall UGen, striking a modal resonant body. Each hit excites up to 16 modes with the ratios of a circular membrane, a square plate, a free bar or a bell.

In Caosmo. When you play the note the ball drops from *Height*. Each bounce loses energy (*Bounce Loss*) and lands sooner, so the hits speed up until the ball comes to rest: a roll that closes on its own. A new note strikes the same body, which keeps ringing.

Parameter	Range (default)	What it does
Gravity	0.5–40 (6)	Gravity. High: fast bounces. Low: slow, spaced hits.
Bounce Loss	0–1 (0.35)	Energy lost per bounce: how long the roll lasts.

Parameter	Range (default)	What it does
Height	0–1 (0.5)	Initial height: strength of the first hit and time until it.
Scatter	0–1 (0.1)	Irregularity of each bounce.
Decay	0.02–8 s (0.8)	Decay of the modes.
Strike	0–1 (0.5)	Brightness of the hit: low excites the low modes; high, the upper ones.
Tension	0–1 (0)	Stretches the mode ratios: more inharmonic, tighter bodies.
Modes	2–16 (10)	Number of modes.
Body	Circle Membrane, Square Membrane, Free Bar, Bell (Circle Membrane)	Body.

Sweet spots. Free Bar with gravity 14: a marble on a xylophone (*Marble Bar*). Bell with gravity 1.2 and a long Decay: moon strikes (*Low Gravity*). A random trigger routed to Scatter changes every drop.

9

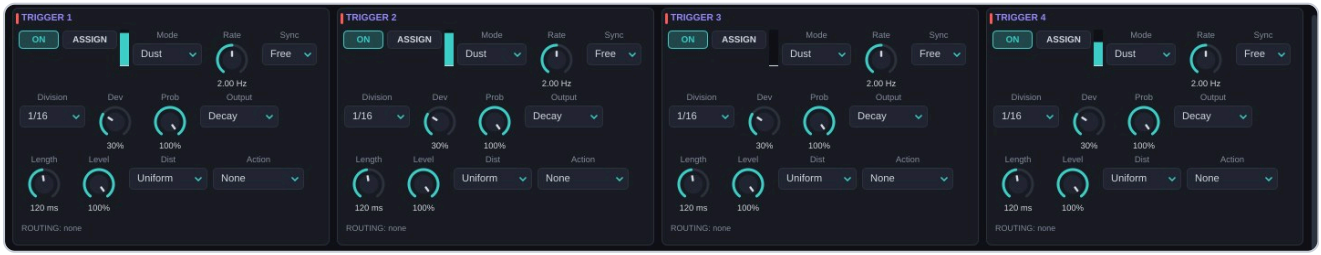
Chance: triggers, distributions and new waves

9.1 Random triggers ×4

Four trigger generators, in the TRIGGER panels at the bottom of the MODULATORS page. They produce events at random moments and can modulate the matrix or act directly on the synth. They come from SuperCollider's Dust, GaussTrig, CoinGate and LorenzTrig UGens.

Mode	How it fires
Dust	Poisson process: Rate events per second on average, with no pattern.
Gauss	A clock whose interval is jittered along a Gaussian bell (Deviation). A human pulse.
Coin	A regular clock; each hit passes only with the given probability.
Lorenz	The instantaneous rate follows a Lorenz attractor: bursts and silences.
Clock	Regular clock with swing (Deviation).

Parameter	Range (default)	What it does
On	On/Off (On)	Enables the generator.
Mode	See table (Dust)	Type of process.
Rate	0.05–50 Hz (2)	Events per second. Modulatable (destination *TrigN Rate*).
Sync / Division	Free, Host / divisions (Free / 1/16)	Rhythm locked to the tempo.
Deviation	0–100 % (30 %)	Gauss deviation, Lorenz depth or Clock swing.
Probability	0–100 % (100 %)	Probability that each trigger passes (CoinGate). Modulatable (*TrigN Prob*).
Output	Gate, Decay, S&H, S&H Decay (Decay)	Signal of the *Trig N* source: gate, exponential decay, held random value or decaying random value.
Length	1–2000 ms (120)	Length of the gate or the decay.
Dist	The 10 distributions (Uniform)	Where the S&H values come from.
Action	None, Retrig Envs, Restart LFOs, Stutter (None)	What the trigger also does: restart the envelopes of the sounding notes, restart the LFOs with Retrig, or open the Deluge stutter while the gate lasts.
Level	0–100 % (100 %)	Source amplitude.



A lamp beside each generator flashes on every trigger.

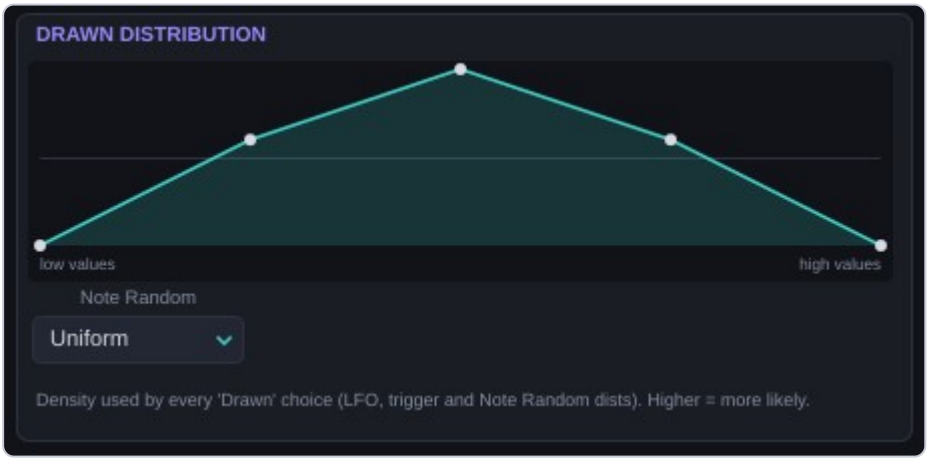
9.2 Distributions

Caosmo's random values no longer have to be uniform. Ten distributions, inspired by SuperCollider's Pwhite, Pgauss, Plprand, Phprand, Pmeanrand, Pexprand, Pcauchy, Pbeta, Ppoisson and Pprob patterns:

Distribution	Shape
Uniform	Every value equally likely.
Gauss	Centred bell: almost always near the middle.
Low / High	Favours low / high values (minimum or maximum of two draws).
Mean	Triangular: mean of two draws.
Exponential	Many small values, few large ones.
Cauchy	Almost always central, with extreme jumps now and then.
Beta	U-shaped: almost always at the extremes.
Poisson	Discrete steps (λ = 4).
Drawn	Whatever you draw in the MATRIX page editor.

They are chosen in three places: each LFO's *Dist* parameter (for Random S&H, Random Smooth and the new waves), each trigger's *Dist*, and *Note Random Dist* on the MATRIX page, which shapes the *Note Random* source.

Drawn distribution. The DRAWN DISTRIBUTION panel on the MATRIX page is a curve editor like the key-track one: draw the probability density (higher means more likely) and Caosmo turns it into a sampling table. It is saved with the preset. By default it is a bell.



9.3 LFO waves with memory

Four new shapes that remember their state from one cycle to the next. *Skew* sets how far they move and *Dist* where the steps come from.

Wave	Behaviour
Brownian	Random walk that bounces off the edges: slow, organic drift.
Geo Brownian	Walk on a logarithmic scale: it moves in proportion to its value, like a price.

Wave	Behaviour
Noise Ring	Wiard's noise ring: a rotating 16-bit register; Skew decides how many bits flip. Low Skew repeats a pattern, high Skew rewrites it.
Markov	Markov chain over 8 levels. Low Skew: steps to neighbouring levels. High: jumps anywhere.

With Loop Lock on, these waves replay from the start on every loop: randomness that repeats exactly.

10

New filters, attractors and effects

10.1 Friction and Spring filters

Type	How it works	Res controls
Friction	A mass on a belt that moves with the signal (Dan Stowell's Friction). It sticks and slips: adds rubbing and squeal. Cutoff is the spring frequency. Louder with a low Cutoff (100–500 Hz).	Dry friction: how much it sticks
Spring	A mass on a non-linear (hardening, Duffing-like) spring driven by the signal: a resonance that rises in pitch with amplitude.	Less damping and more non-linearity

10.2 New attractors

The attractor modulators have seven more types:

Type	P1	P2	Character
Henon	A	B	Hénon map: jumps between points of a fractal curve.
Latoocarfian	A	B	Pickover's map: chaotic steps or cycles.
Standard	K	Init	Chirikov's map: near-periodic with low K, chaotic with high K.
Gingerbread	X0	Y0	Deterministic path with calm regions.
FitzHugh	Current	Eps	A neuron: regular spikes or silence.
Lotka-Volterra	Prey	Pred	Prey-predator cycles: slow rises and sudden falls.
ACT	Alpha	Immig	Three competing species, chaotic.

The first four are maps: they take one step per Rate cycle and Smooth joins the steps.

10.3 Spectral and cycle effects

Six new effects for the bus slots. The first four work on the spectrum (1024-point FFT, 21 ms of internal latency inside the bus), like SuperCollider's PV_ UGens.

Effect	P1	P2	P3	P4	Option
Diffuse	Amount	Rate (0.05–20 Hz)	Level	—	Live / Freeze
Mag Noise	Amount	Smooth	Level	—	—
Rand Comb	Density	Rate (0.05–20 Hz)	Smear	Level	Live / Freeze
Noise Synth	Thresh	Memory (1–16 frames)	Level	—	Unstable / Stable
Wave Loss	Drop	Burst (1–16 cycles)	Level	—	Random / Regular
Squiz	Ratio (1–10×)	Cycles (1–10)	Level	—	—

- **Diffuse** (PV_Diffuser): scrambles the phase of each band; *Rate* changes the scrambling over time. *Freeze* holds the spectrum with random phases: an infinite sustain.

- **Mag Noise** (PV_MagNoise): multiplies each band's amplitude by noise; *Smooth* makes it slow and undulating.
- **Rand Comb** (PV_RandComb): switches bands off at random; *Density* is the proportion that passes, *Smear* softens the changes. *Freeze* fixes the selection.
- **Noise Synth** (PV_NoiseSynth): separates stable bands (tones) from unstable ones (noise). *Unstable* keeps only the noise; *Stable*, only the tones.
- **Wave Loss**: drops whole wave cycles between zero crossings; *Drop* is the proportion, *Burst* how many cycles in a row.
- **Squiz**: replays each group of cycles faster, "squeezing" it upwards: a raw, grainy pitch shifter.

So the level does not jump when you switch effect, Level at 62.5 % is unity gain in the four spectral effects.

11

Filter, envelope, EQ and output

11.1 FILTER

Each lane has a stereo, per-voice filter with fourteen types. The state-variable filters (SVF, Simper's design) stay stable even when the cutoff is modulated at audio rate.

Type	How it works	Res controls
LPF	Cascaded SVF low-pass	Resonance of the last stage
HPF	Cascaded SVF high-pass	Resonance
BPF	Unity-gain band-pass	Narrowness of the band
Notch	Band reject	Narrowness
Peak	Bell at the cutoff frequency	Gain, up to +18 dB
Low Shelf	Bass shelf	Gain, up to +18 dB
High Shelf	Treble shelf	Gain, up to +18 dB
All-Pass	Changes phase without changing level; useful when modulated	Sharpness of the phase turn
Ladder	4-pole ladder (24 dB/oct) with input saturation, transistor style	Feedback up to self-oscillation
Comb	Comb filter: a 1/cutoff delay with feedback. Tuned to Cutoff, not to the note	Feedback (up to 0.97)
Formant	Three bands at vowel frequencies; Cutoff sweeps A → E → I → O → U	Narrowness of the formants
Phaser	Four staggered all-passes mixed with the dry signal	Feedback
Friction	A mass on a belt that sticks and slips (see *Friction and Spring filters*)	Dry friction
Spring	A mass on a hardening non-linear spring	Less damping, more non-linearity

Parameter	Range (default)	What it does
Filter Type	The 14 types (LPF)	Type.
Filter Slope	12, 24, 36, 48 dB (12)	Number of stages for LPF, HPF, BPF, Notch and All-Pass. The other types have a fixed slope.
Cutoff	20 Hz – 20 kHz (20 kHz)	Cutoff or centre. Per-note modulatable.
Res	0–100 % (10 %)	See the table above.
LPG Mode	Amp, Amp + Filt (Amp)	*Amp + Filt*: the envelope also opens the filter, like a Buchla *low pass gate*. The actual cutoff goes from 20 Hz (envelope at 0) up to Cutoff (envelope at 1).

To make the filter track the note, create a *Keytrack → Cutoff* route. Comb with that keytrack route and high Res turns any unpitched engine into a plucked string.

11.2 ENV GEN

One envelope per lane and per note controls the volume (and the filter in LPG mode). Its output is also the matrix source *Lane Env*. It has four modes:

ADSR. A classic envelope with delay and per-segment curves.

Vactrol. Simulates an opto-coupler (a lamp facing a photoresistor), the heart of *low pass gates*. It has two components, fast and slow, each with its own rise and fall. **Memory** imitates a real photoresistor: the more it has been lit recently, the longer it takes to fade (up to five times longer). Play fast notes and the tails stretch by themselves.

LFO. The envelope is a per-note LFO starting at phase 0: tremolos, pulses, free-running rhythmic *gates*.

VENV. The envelope is one of the four drawn shapes from the MODULATORS page, with its own time for this lane.

Parameter	Mode	Range (default)	What it does
Env Mode	—	ADSR, Vactrol, LFO, VENV (ADSR)	Mode.
Env Sub Mode	All	Sustain, One-Shot, Sustain-Loop (Sustain)	*One-Shot*: ignores the end of the note (percussion). *Sustain-Loop*: in VENV mode, loops the sustain zone.
Delay	ADSR	0–2000 ms (0)	Wait before the attack.
Attack	ADSR	0.3 ms – 10 s (5 ms)	Rise. It starts from the current level, so repeated notes don't click.
Decay	ADSR	1 ms – 10 s (300 ms)	Fall to Sustain.
Sustain	ADSR	0–100 % (80 %)	Level while the note is held.
Release	ADSR	1 ms – 15 s (400 ms)	Tail after release.
Att / Dec / Rel Curve	ADSR	Lin, Exp, Log (Lin / Exp / Exp)	Shape of each segment. *Exp* sounds natural on falls.
Rise Fast / Fall Fast	Vactrol	0.5–500 ms (5) / 1–2000 ms (80)	Fast component.
Rise Slow / Fall Slow	Vactrol	5–2000 ms (120) / 20–8000 ms (900)	Slow component: the tail.
Vac Mix	Vactrol	0–100 % (50 %)	Weight of the slow component.
Memory	Vactrol	0–100 % (30 %)	Memory effect.
Vac Shape	Vactrol	0–100 % (40 %)	Curvature of the lamp response: higher gives more "pop" and thinner tails.
Vac Drift	Vactrol	0–100 % (10 %)	Random per-note timing variation, up to ±35 %.
Env LFO Rate	LFO	0.05–40 Hz (2)	Speed.
Env LFO Shape	LFO	Sine, Triangle, Saw Down, Square, Ramp Up (Sine)	Shape. *Saw Down* gives percussive pulses.
Env VENV Shape	VENV	VENV 1–4 (VENV 1)	Shape used.
Env VENV Time	VENV	0.01–30 s (1)	Length of the shape on this lane (independent of the VENV's own Time).

MIDI velocity always scales the level linearly. The matrix source *Velocity* lets you send it to any other destination as well.

11.3 Lane EQ

A five-band stereo equaliser on the lane bus: high-pass, low shelf, mid bell, high shelf and low-pass. Drag the nodes on the curve; the wheel changes the mid band's Q or the filters' slope. **EDIT** opens a large editor.

Parameter	Range (default)
EQ On	On/Off (On)
EQ HPF / HPF Slope	20 Hz – 2 kHz (20) / 12–48 dB (12)
EQ Low Freq / Low Gain	20 Hz – 1 kHz (120) / ±18 dB (0)
EQ Mid Freq / Mid Gain / Mid Q	100 Hz – 10 kHz (1 kHz) / ±18 dB (0) / 0.2–8 (1)
EQ High Freq / High Gain	1–18 kHz (6 kHz) / ±18 dB (0)
EQ LPF / LPF Slope	200 Hz – 20 kHz (20 kHz) / 12–48 dB (12)

Snapshots and morph

The EQ stores up to **four curves** (snapshots A–D):

1. Set up a curve on A.
2. Select B: the first time, B copies the current curve. Edit it.
3. Repeat with C and D if you like.
4. **EQ Morph** (0–100 %) travels through the used snapshots in the order A → B → C → D, interpolating frequencies logarithmically and gains linearly.

Selecting a snapshot moves the morph onto it, so you hear what you edit. **RST** returns to a single flat snapshot. EQ Morph is a modulation destination (for global sources): a slow LFO on it gives several bands of filtering moving together.

11.4 OUTPUT

Parameter	Range (default)	What it does
Level	−60 to +12 dB (0)	Lane volume. Per-note modulatable.
Pan	−1 to +1 (0)	Lane pan.
Width	0–2 (1)	M/S width: 0 mono, 1 natural, 2 double sides.
Voice Spread	−1 to +1 (0)	Places each voice at a fixed stereo position according to note order (left, right, centre-left…). The sign flips the sides. A chord opens up like a section.
Bass Mono	0–500 Hz (0 = off)	Below this frequency the lane is mono.
Drive	0–100 % (0)	Saturation on the lane bus, with up to ×15 gain and level compensation.
Drive Flavour	Tape, Clip, Fold, Transformer, Wavefolder (Tape)	See table.
FX1 / FX2 / FX3 Send	−60 to 0 dB (−60 = off)	Send to buses A, B and C (post-fader).

Flavour	Character
Tape	Asymmetric hyperbolic tangent: even harmonics, warm, gentle compression.
Clip	Hard clipping: aggressive, digital.
Fold	Sine folding: as you push it the wave folds over itself and metallic harmonics appear.
Transformer	Saturation with a transformer's rounded "shoulder": dense, odd harmonics.
Wavefolder	Multiple triangle folding, West Coast style: very bright, timbre changes with input level.

Pan, Width, Drive, the sends and EQ Morph act on the lane bus, so they only accept modulation from global sources. The matrix marks a route that cannot have any effect in red.

12

Modulation

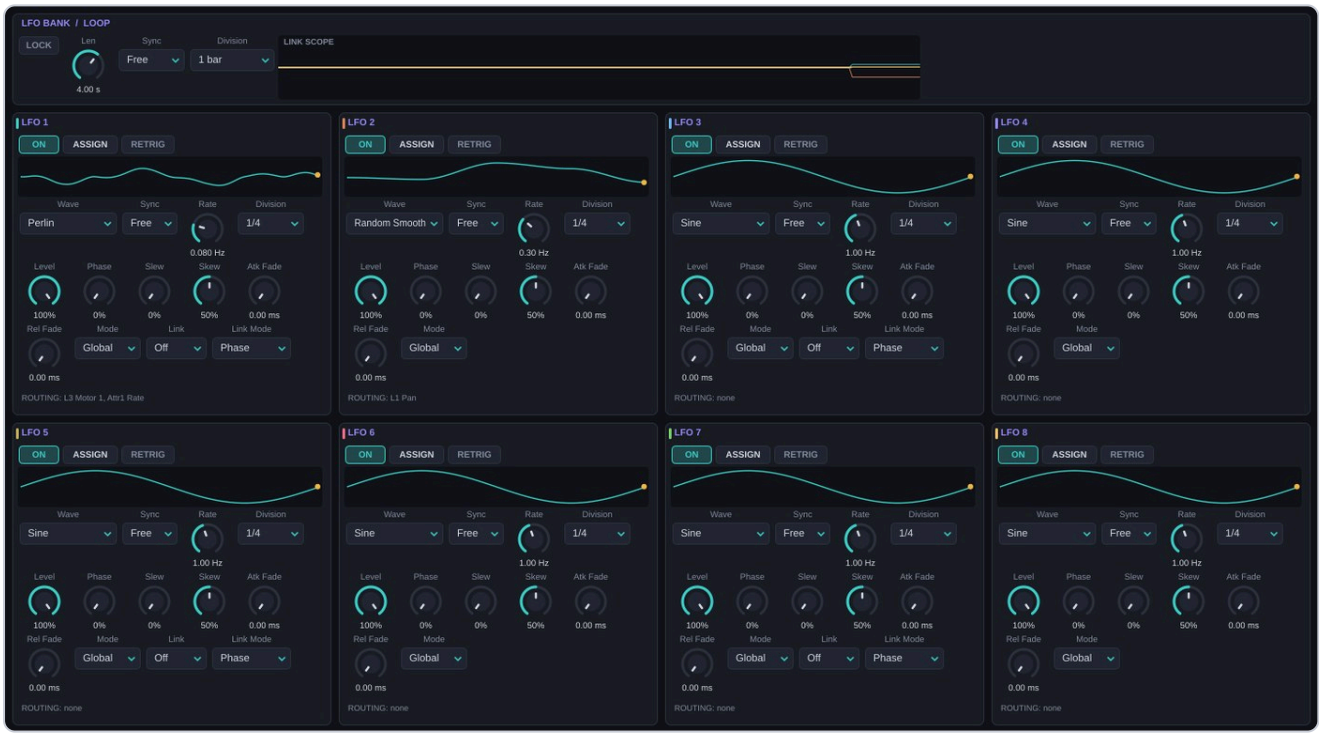
Caosmo has 16 modulators and 24 routes. There are two ways to connect them:

- **ASSIGN**: press ASSIGN on a modulator and click any control. Esc cancels.
- **Right-click → Modulate with…** on the destination control.

Both create a route in the matrix, where you can fine-tune its amount, curve and behaviour.

Global and Poly. A *Global* modulator is a single instance shared by every note: ideal for ensemble movement and for lane-bus or master destinations. A *Poly* modulator has an independent copy per note, starting with the note: every voice moves on its own.

12.1 LFO ×8



Parameter	Range (default)	What it does
On	On/Off (On)	Enables the LFO.
Wave	Sine, Triangle, Saw Up, Saw Down, Pulse, Random S&H, Random Smooth, Perlin, Brownian, Geo Brownian, Noise Ring, Markov (Sine)	Shape. *Random S&H*: random steps. *Random Smooth*: interpolated random. *Perlin*: two-octave gradient noise, organic. The last four have memory (see *LFO waves with memory*).
Rate	0.01–50 Hz (1)	Free-running speed. Modulatable.
Sync	Free, Host (Free)	*Host*: synced to the DAW position (in Global mode with the transport running it follows the ruler).
Division	8 bars … 1/32, 1/4T, 1/8T, 1/16T, 1/4D, 1/8D (1/4)	Length of one cycle in Host mode. Rate modulation multiplies it by powers of 2.

Parameter	Range (default)	What it does
Level	0–100 % (100 %)	Amplitude. Modulatable.
Phase	0–100 % (0)	Start phase (and offset in Host mode).
Slew	0–100 % (0)	Smooths the jumps of Pulse and S&H.
Skew	0–100 % (50 %)	Moves the midpoint of the cycle: a triangle becomes a saw; a pulse changes its width.
Atk Fade	0–5000 ms (0)	Gradual fade-in of the LFO after the note starts.
Rel Fade	0–5000 ms (0)	Gradual fade-out after release.
Mode	Global, Poly (Global)	One instance or one per note.
Retrig	On/Off (Off)	Restarts the phase on every note.
Dist	The 10 distributions (Uniform)	Distribution of the values of Random S&H, Random Smooth and the waves with memory.

Links (LINK)

LFOs 1, 3, 5 and 7 can lead the following ones. Followers take the leader's speed but keep their own wave, Level and Skew.

Link	Group
Dual (LFO 1, 3, 5, 7)	The LFO and the next one: 1–2, 3–4, 5–6, 7–8.
Quad (LFO 1 and 5)	Four LFOs: 1–4 or 5–8.
Oct (LFO 1)	All eight.

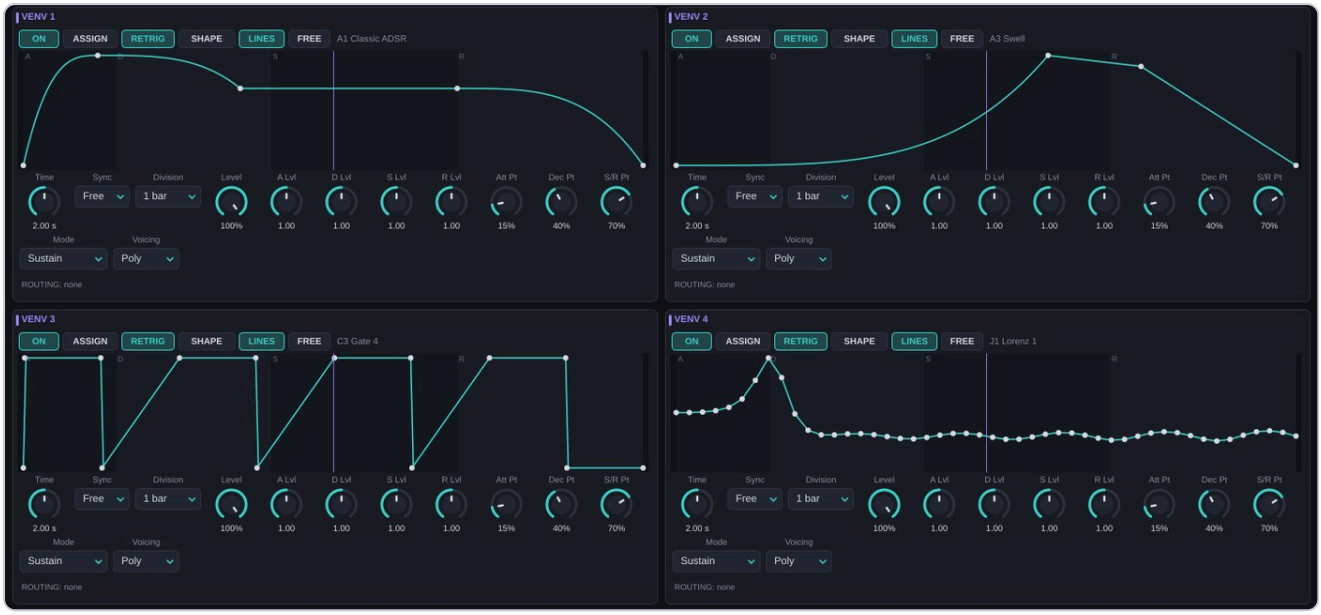
Link Mode	Relationship of each follower k (1, 2, 3…)
Sync	Same phase as the leader.
Phase	Spread phases: k/n of a cycle offset. Ideal for fanned movement and circular panning.
Divide	Leader speed ÷ (k+1): 1:2:3:4 polyrhythms.
Pendulum	Odd followers run each cycle backwards.
Mirror	Odd followers invert their output.

Loop Lock

Random shapes normally never repeat. **Loop Lock** restarts their generator at a set interval, so the same random sequence comes back again and again: randomness shaped like a *riff*.

Parameter	Range (default)
LFO Loop Lock	On/Off (Off)
LFO Loop Len	0.25–16 s (4); modulatable
LFO Loop Sync	Free, Host (Free)
LFO Loop Division	1/4 bar, 1/2 bar, 1/2T bar, 3/4 bar, 1 bar, 1T bar, 1.5 bars, 2, 3, 4, 6 or 8 bars (1 bar)

12.2 VENV ×4



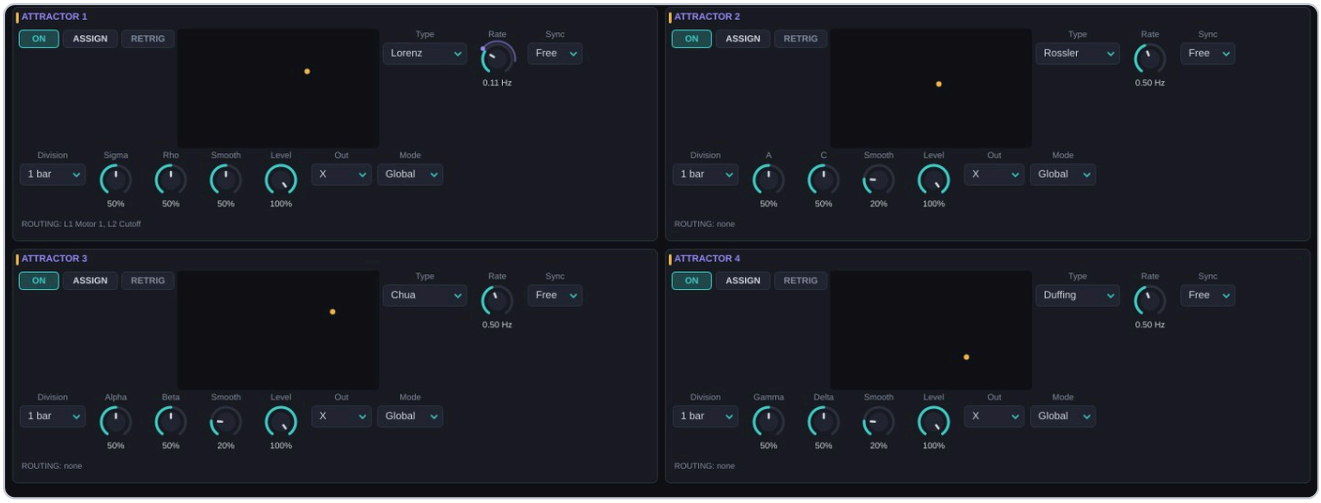
VENVs are drawable envelopes: a free shape travelled over a given time, with attack, decay, sustain and release zones.

Editor. In **LINES** mode you drag points; double-click adds one; right-click deletes it or changes the segment's curvature; Alt snaps to the grid; the wheel zooms; Shift + drag pans the view. In **FREE** mode you draw freehand. The shape menu has **100 factory shapes** in ten banks (A–J) plus your saved shapes.

Parameter	Range (default)	What it does
On	On/Off (On)	Enables the VENV.
Time	0.01–30 s (2)	Total length. Modulatable.
Sync / Division	Free, Host / LFO divisions (Free / 1 bar)	Length in musical time.
Level	0–100 % (100 %)	Amplitude. Modulatable.
Attack / Decay / Sustain / Release Lvl	0–2 (1)	Scales each zone without redrawing.
Attack Pt / Decay Pt / Sus-Rel Pt	0–100 % (15 / 40 / 70 %)	Zone boundaries on the shape.
Mode	1-Shot, Sustain, Sustain-Loop (Sustain)	*1-Shot*: plays the whole shape. *Sustain*: stops at the start of the Sus-Rel zone until release. *Sustain-Loop*: loops the sustain zone (yellow band) while the note is held.
Voicing	Global, Poly (Poly)	Per note or shared.
Retrig	On/Off (On)	Restarts on every note.

A long 1-Shot VENV on a chaos parameter is the most direct way to write a sound's "story".

12.3 Attractors ×4



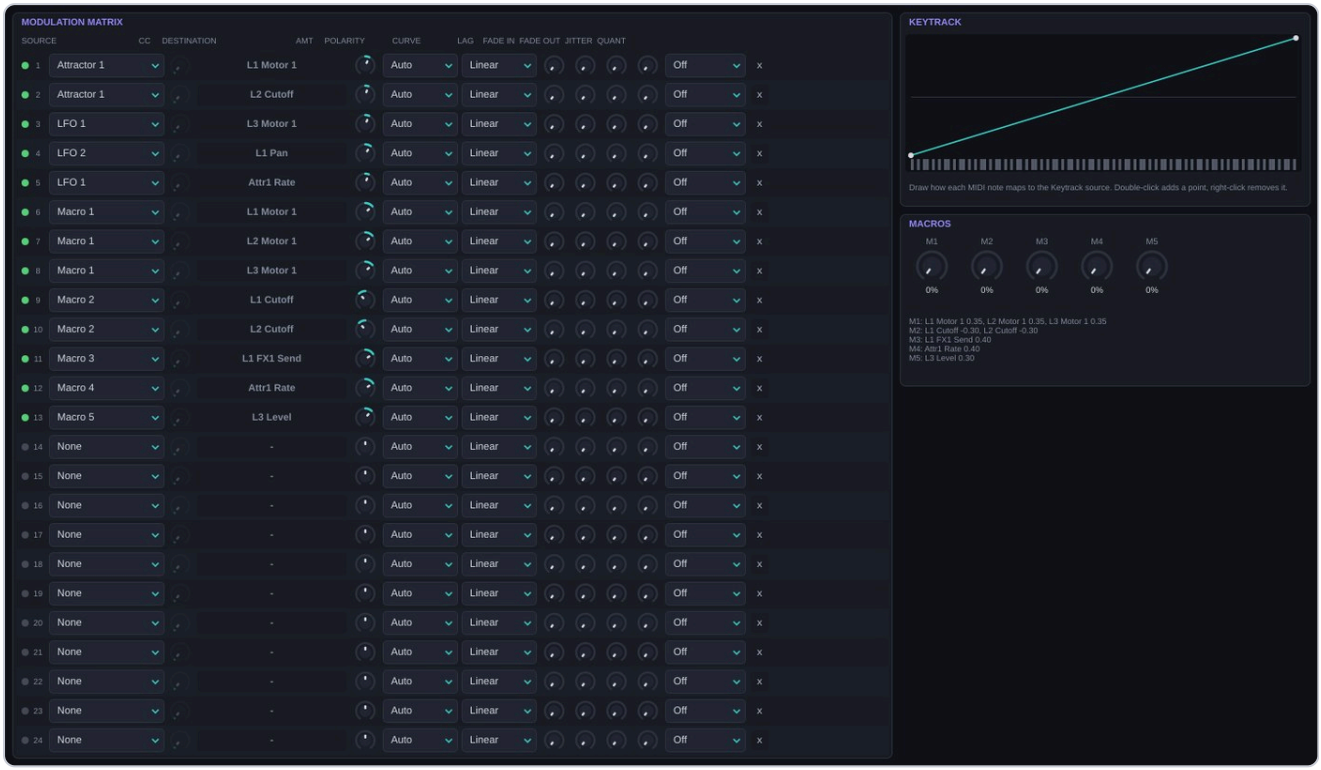
Modulators that follow the trajectory of a strange attractor: smooth, never repeating, with correlation between their three coordinates. The rotating 3-D portrait shows the orbit.

Type	P1	P2	Character
Lorenz	σ 5–20	ρ 20–45	Two lobes: the signal jumps unpredictably between two zones.
Rossler	a 0.1–0.3	c 4–9	An almost periodic spiral with occasional leaps.
Chua	α 8–11	β 12–18	Double scroll: alternates between two centres.
Duffing	γ 0.2–0.6	δ 0.1–0.4	Forced oscillation with well jumps; Z follows the force.
Sprott B	a 0.6–1.4	b 0.5–1.5	A smooth, symmetric attractor.
Henon ... ACT			Seven new types in version 0.5: see *New attractors*.

Parameter	Range (default)	What it does
On	On/Off (On)	Enables the attractor.
Type	The 12 types (1 Lorenz, 2 Rossler, 3 Chua, 4 Duffing)	System.
Rate	0.01–20 Hz (0.5)	Speed. Modulatable.
Sync / Division	Free, Host / divisions (Free / 1 bar)	Speed tied to tempo.
P1 / P2	0–100 % (50 %)	Coefficients (see table).
Smooth	0–100 % (20 %)	Output smoothing.
Level	0–100 % (100 %)	Amplitude. Modulatable.
Out	X, Y, Z, Blend (X)	Coordinate used.
Mode	Global, Poly (Global)	In Poly, each note has its own attractor with its own seed.
Retrig	On/Off (Off)	Restarts on every note.

An attractor modulating the Rate of another attractor (or of an LFO) creates coupled systems: the *Weather Systems* category is built that way.

12.4 Matrix ×24



Each route takes a source to a destination with its own processing. The **x** at the end of the row deletes the route; the row's **CC** button learns a controller for that route.

Parameter	Range (default)	What it does
Source	37 sources (None)	Where the modulation comes from.
Dest	Grouped destinations (None)	The parameter being modulated.
Amount	−1 to +1 (0)	Amount. ±1 covers the destination's full range.
Polarity	Auto, Unipolar, Bipolar (Auto)	*Auto* respects the source's nature (LFOs, attractors, keytrack, pitch bend and Note Random are bipolar; the rest are unipolar). *Unipolar* / *Bipolar* convert it.
Curve	Linear, Exp, Log, S-Curve (Linear)	Response. *Exp*: more detail at low values (ideal for velocity). *Log*: faster at the start. *S-Curve*: softens the extremes.
Lag	0–1000 ms (0)	Smoothing: turns steps into ramps.
Fade In	0–5000 ms (0)	The route fades in after the note starts (delayed vibrato).
Fade Out	0–5000 ms (0)	The route fades out after release.
Jitter	0–100 % (0)	Adds smoothed random variation.
Quant	Off, Semitone, Major, Minor, Penta, Chromatic (Off)	Lane *Pitch* destinations only: rounds the modulation to semitones or to scale intervals counted from the note played.
CC	0–127 (1)	Controller number when the source is *CC*.

Sources

Group	Sources	Per note
LFOs	LFO 1–8	When in Poly

Group	Sources	Per note
VENV	VENV 1–4	When in Poly
Attractors	Attractor 1–4	When in Poly
Envelopes	Lane 1–3 Env	Yes
MIDI	Velocity, Release Vel, Keytrack, Mod Wheel, Aftertouch, Pitch Bend, Expression (CC 11), Sustain (CC 64), CC	Velocity, Release Vel and Keytrack
MPE	MPE Pressure, MPE Slide (CC 74), MPE Bend	Yes
Macros	Macro 1–5	No
Triggers	Trig 1–4	No
Other	Note Random (a fixed random value per note, shaped by *Note Random Dist*)	Yes

Destinations

Group	Destinations
Lane 1–3 (per note)	Pitch, Fine, Cutoff, Res, Level, V.Spread, Sub1 Lev, Sub2 Lev, Motor 1–6, Env Att, Env Dec, Env Sus, Env Rel
Lane 1–3 (global only)	Pan, Width, Drive, FX1–3 Send, EQ Morph
Modulators	LFO 1–8 Rate and Level, VENV 1–4 Time and Level, Attr 1–4 Rate and Level
Triggers	Trig 1–4 Rate and Prob
Global	LFO Loop Len, Glide
Master	Master Low Gain, Mid Gain, High Gain, Sat Drive, Sat Mix, Comp Thresh, Master Width, Output Gain
FX	FX A/B/C Level and P1–P5 of each of the 9 slots

Status lights

Light	Meaning
Green	The route is active.
Blue	Modulation loop: a modulator modulates itself through others. It works, but it can become unstable.
Amber	The destination is a *Motor N* the current engine doesn't have (e.g. Motor 5 on Chua).
Red	Per-note source to a global destination: the route has no effect.

12.5 Keytrack

A drawable curve that turns the MIDI note (0 to 127) into a modulation value. Double-click adds a point and right-click removes it. The *Keytrack* source outputs that value. By default it is a straight line, but you can make only high notes modulate, keep the middle of the keyboard neutral, or give each zone different behaviour. The *Strata* category uses curves that make high notes unstable.

12.6 Macros

Five global controls (0–100 %) on the bottom bar (M1–M5), meant for playing and automating. *Right-click a control → Send to Macro* creates the route directly. The **MACROS** view on the MATRIX page shows what each one moves. Every factory preset has all five macros assigned to something audible.

12.7 MIDI, MIDI Learn and MPE

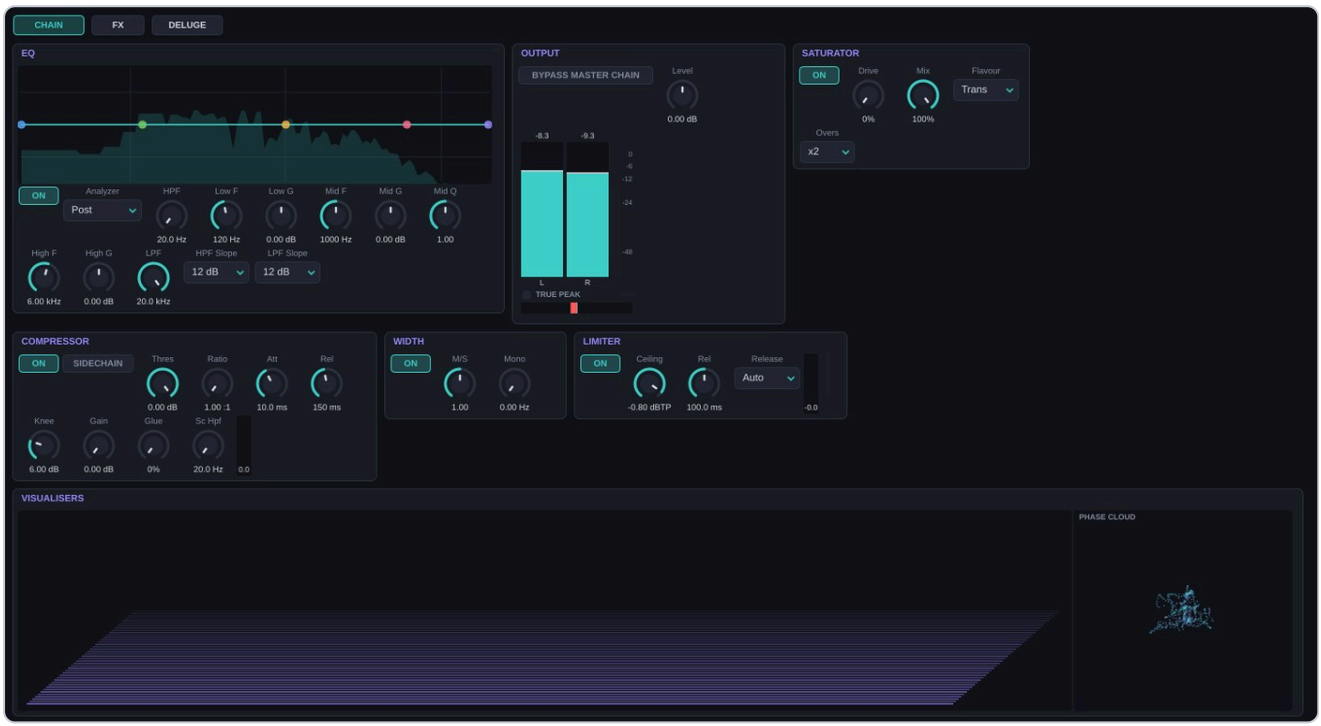
- **MIDI Learn**: right-click a control → *MIDI / MPE → MIDI Learn* and move a controller. Caosmo creates a *CC* route in the matrix, so it is saved with the preset and you can adjust its amount and curve.
- **Per-lane channel**: in LANE, for splits or three parts from three MIDI tracks.
- **PB Range**: global pitch-bend range, 0–24 semitones (2).
- **MPE**: choose *Lower* (master channel 1, notes on 2–16) or *Upper* (master channel 16, notes on 1–15). Each note gets its own bend (*MPE PB Range*, 0–96 semitones, 48 by default), pressure and slide (CC 74), available as per-note sources. With MPE on, lane channels are ignored.

In REAPER, for MPE set the track's MIDI input to *All channels* and don't force a channel on the send to the plugin.

13

Master

The master chain processes the final sum: **EQ → Saturator → Compressor → Width → Limiter → Output**.



Parameter	Range (default)	What it does
Bypass Master Chain	On/Off (Off)	Skips EQ, saturator, compressor and width. Limiter and Output stay active.
Master EQ On and bands	Same as the lane EQ	Five-band EQ on the mix.
Analyzer	Off, Pre, Post, Pre+Post (Post)	Spectrum before and/or after the EQ, drawn under the curve.
Sat On	On/Off (On)	Enables the saturator.
Sat Drive	0–100 % (0)	Gain into the saturation, up to ×21, with compensation.
Sat Mix	0–100 % (100 %)	Dry/saturated mix (parallel).
Sat Flavour	Trans, Tube (Trans)	*Trans*: firm symmetric curve, odd harmonics. *Tube*: biased curve, even harmonics, warmer.
Sat Oversampling	Off, x2, x4 (x2)	Oversampling that reduces saturation aliasing.
Comp On	On/Off (On)	Enables the compressor. With Ratio 1:1 and Makeup 0 it does nothing.
Comp Thresh	−60 to 0 dB (0)	Threshold. Modulatable.
Comp Ratio	1:1 – 20:1 (1:1)	Ratio.
Comp Attack / Release	0.1–200 ms (10) / 10–2000 ms (150)	Times.
Comp Knee	0–24 dB (6)	Soft knee.

Parameter	Range (default)	What it does
Comp Makeup	0–24 dB (0)	Makeup gain.
Comp Glue	0–100 % (0)	Blends detection from the input (*feed-forward*, precise) with detection from the output (*feedback*, like classic bus compressors): more glue, softer, "stickier" compression.
Comp SC HPF	20–500 Hz (20)	High-pass on the detector: lows stop pumping the compressor.
Comp Ext Sidechain	On/Off (Off)	The detector listens to the plugin's sidechain input. Enable it and route audio to it in the DAW.
Width On / Master Width	On / 0–2 (1)	M/S width.
Master Bass Mono	0–500 Hz (0 = off)	Lows in mono below this frequency.
Limiter On	On/Off (Off)	Limiter with 1.5 ms look-ahead and inter-sample *true peak* estimation.
Limiter Ceiling	−12 to 0 dBTP (−1)	Ceiling.
Limiter Release	1–1000 ms (100)	Release in Manual mode.
Limiter Release Mode	Manual, Auto (Auto)	*Auto*: from 30 ms to about 630 ms depending on average reduction; transparent on dense material.
Output Gain	−60 to +12 dB (0)	Final volume. Modulatable.

Latency. The plugin always reports 1.5 ms of latency (the look-ahead), whether the limiter is on or not. That way the DAW compensates without jumps when you switch it.

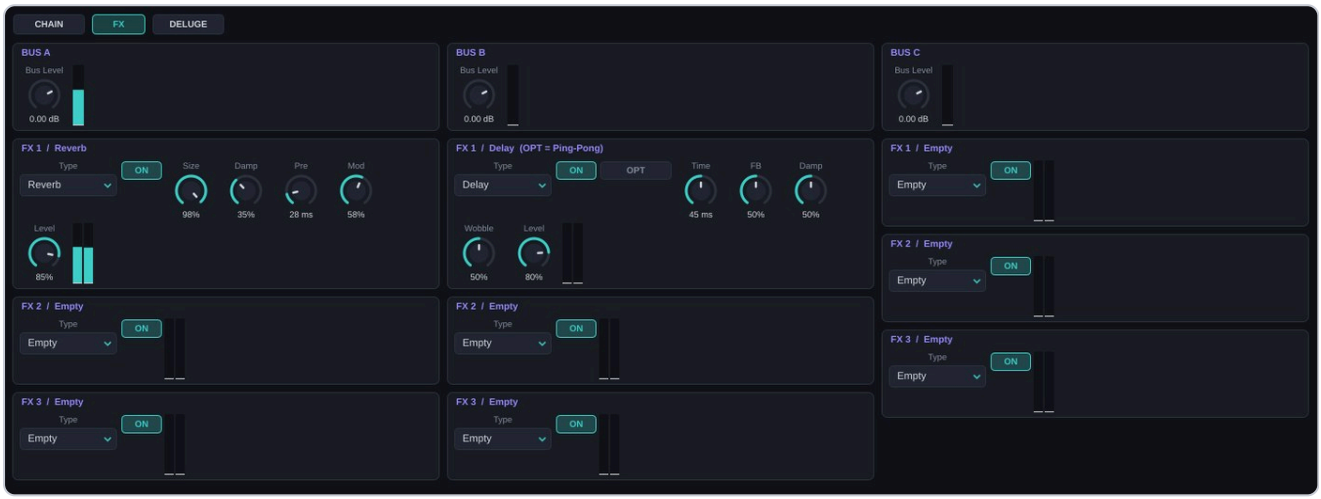
Meters. Peak, true peak, compressor and limiter gain reduction, and stereo correlation (+1 mono, 0 decorrelated, −1 out of phase).

Visualisers. Spectral waterfall, phase cloud, terrain and galaxy. Click to change view, wheel to zoom and right-click to change palette. They are purely visual.

14

Effect buses

Three buses (A, B and C) fed by the lane sends. Each bus has three slots in series and returns its signal 100 % wet to the mix, before the Deluge rack and the master. By default bus A holds a Reverb and bus B a Delay.



Parameter	Range (default)	What it does
Bus Level	−60 to +6 dB (0)	Return level. Modulatable.
Slot Type	Empty, Reverb, Delay, Freq Shift, Granular, Howl, Ring Mod, Diffuse, Mag Noise, Rand Comb, Noise Synth, Wave Loss, Squiz	Effect. The last six are described in *Spectral and cycle effects*.
Slot On	On/Off (On)	Enables the slot.
P1–P5	0–100 % (50 %; P5 80 %)	Effect-dependent parameters. All modulatable.
Option	On/Off (Off)	Delay: Stereo / Ping-Pong. Granular, Diffuse and Rand Comb: Live / Freeze. Noise Synth: Unstable / Stable. Wave Loss: Random / Regular.

Effect	P1	P2	P3	P4	P5
Reverb	Size	Damp	Pre (0–200 ms)	Mod	Level
Delay	Time (1 ms – 2 s)	FB (0–98 %)	Damp	Wobble	Level
Freq Shift	Shift (±2 kHz)	FB	Level	—	—
Granular	Size (10–500 ms)	Density (1–40 grains/s)	Scatter	Pitch (±24 st)	Level
Howl	Freq (40 Hz – 4 kHz)	FB (0–120 %)	Damp	Drive	Level
Ring Mod	Freq (1 Hz – 4 kHz)	Level	—	—	—

- **Reverb**: a feedback delay network with Hadamard mixing, pre-delay and diffusers. *Mod* adds movement to the lines (chorus in the tail).
- **Delay**: stereo or ping-pong, with a filter in the feedback (*Damp*) and *Wobble*, a slow tape-like time drift.
- **Freq Shift**: Hilbert-transform frequency shifter. It moves every frequency by the same amount (inharmonic). With FB it makes rising or falling spirals.

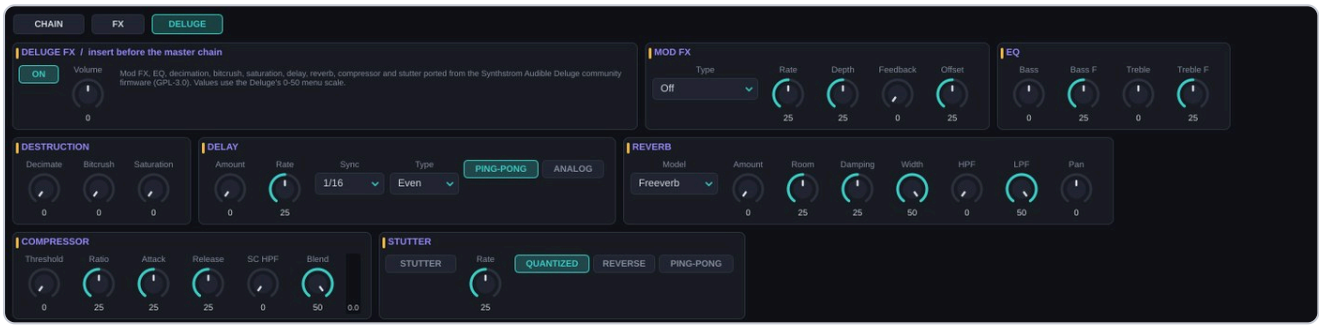
- **Granular**: records the input into a buffer and plays it back in grains. *Freeze* freezes the buffer.
- **Howl**: a delay tuned to *Freq* with a band-pass in the feedback: mic-and-speaker style howl. Above 100 % FB it can self-oscillate; *Drive* saturates it.
- **Ring Mod**: multiplies the signal by a sine: from tremolo (low frequencies) to inharmonic bells.

15

DELUGE FX

The DELUGE FX rack holds the effects from the **Synthstrom Audible Deluge** community firmware, copied from its open-source repository and compiled as is: the same 32-bit fixed-point arithmetic, the same 64-sample blocks and the same tables. It is an insert placed between the sum of lanes and FX and the master chain.

The controls use the Deluge's **0–50 menu scale** (−25 to +25 for bipolar ones) and go through the firmware's own conversions. With everything at its neutral position the level does not change.



The internal order is the Deluge's: stutter → Mod FX → EQ → delay → decimation and bitcrush → saturation → reverb → compressor.

15.1 Mod FX

Parameter	Range (default)	What it does
Mod FX	Off, Flanger, Chorus, Phaser, Stereo Chorus, Warble, Dimension, Grain (Off)	Type.
Rate	0–50 (25)	Speed of the internal LFO. In Grain: grain rate.
Depth	0–50 (25)	Depth. In Grain: mix. **At 0 the Mod FX is off** (as on the Deluge), except Grain.
Feedback	0–50 (0)	Feedback (flanger, phaser). In Grain: pitch randomness.
Offset	0–50 (25)	Base delay of flanger/chorus or phaser centre. In Grain: density.

Warble is a chorus with irregular, tape-like modulation. *Dimension* is inspired by the Roland Dimension D stereo chorus. *Grain* is the Deluge's granulator, which feeds the reverb.

15.2 EQ and destruction

Parameter	Range (default)	What it does
Bass / Treble	−25 to +25 (0)	Bass and treble shelves.
Bass Freq / Treble Freq	0–50 (25)	Frequency of each shelf.
Decimation	0–50 (0)	Sample-rate reduction (deliberate aliasing).

Parameter	Range (default)	What it does
Bitcrush	0–50 (0)	Resolution reduction.
Saturation	0–15 (0)	The Deluge's digital saturation.

15.3 Delay

Parameter	Range (default)	What it does
Amount	0–50 (0)	Echo feedback and level. 0 = off.
Rate	0–50 (25)	Without sync: the time. With sync: multiplies the synced time (25 = exact).
Sync	Off, 2 bars, 1 bar, 1/2, 1/4, 1/8, 1/16, 1/32, 1/64, 1/128 (1/16)	Time locked to the DAW tempo.
Sync Type	Even, Triplet, Dotted (Even)	Note value.
Ping-Pong	On/Off (On)	Stereo bounce.
Analog	On/Off (Off)	Analogue mode: saturation and the response of a *bucket brigade* (BBD) delay via an impulse response.

15.4 Reverb

Parameter	Range (default)	What it does
Model	Freeverb, Mutable, Digital (Freeverb)	*Freeverb*: Jezar's classic, dense and bright. *Mutable*: derived from the reverb in Mutable Instruments modules, diffuse and modulated. *Digital*: the Deluge's own algorithm.
Amount	0–50 (0)	Reverb send. 0 = off.
Room / Damping / Width	0–50 (25 / 25 / 50)	Size, high-frequency absorption and width.
HPF / LPF	0–50 (0 / 50)	Tail filters.
Pan	−25 to +25 (0)	Reverb position.

15.5 Compressor

The Deluge's master compressor: RMS detection with feedback, very musical when pumping.

Parameter	Range (default)	What it does
Threshold	0–50 (0)	Amount of compression. 0 = off.
Ratio	0–50 (25)	Ratio.
Attack / Release	0–50 (25)	Times.
Sidechain HPF	0–50 (0)	High-pass on the detector.
Blend	0–50 (50)	Parallel compression: 50 = fully compressed.

The tab has its own gain-reduction meter.

15.6 Stutter and volume

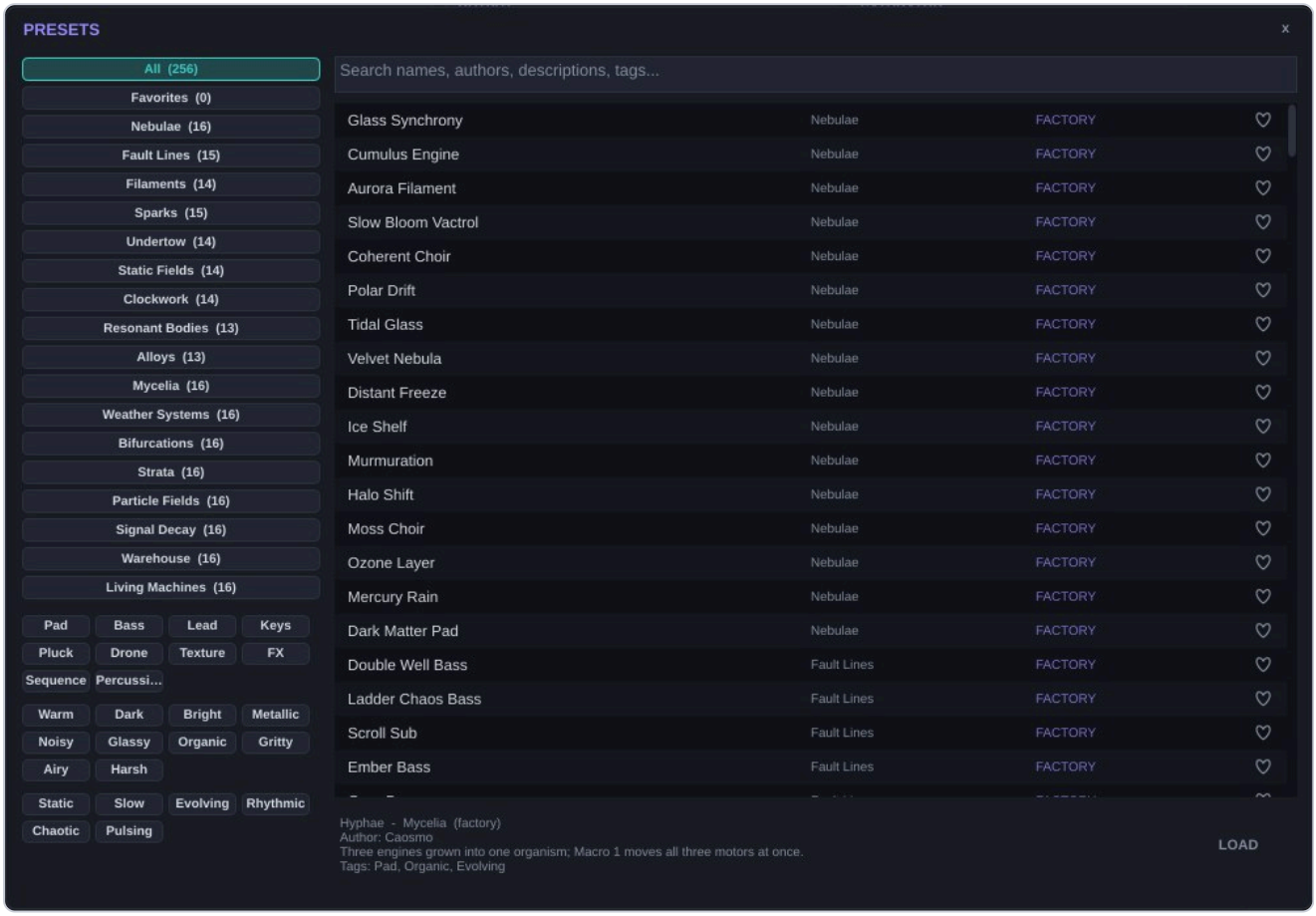
Parameter	Range (default)	What it does
Stutter	On/Off (Off)	While on, repeats the fragment that has just played (*beat repeat*). Meant to be automated or mapped to a MIDI button.
Stutter Rate	0–50 (25)	Repeat length. It is fixed when the stutter starts.
Quantized	On/Off (On)	Snaps the length to the tempo grid.
Reversed	On/Off (Off)	Repeats backwards.
Ping-Pong	On/Off (Off)	Alternates forwards and backwards.
Volume	−25 to +25 (0)	Output level of the rack.

Tip. Automate *Deluge Stutter* in REAPER with a short automation item on the last beat of a bar: fill-style repeats.

16

Presets

16.1 The browser



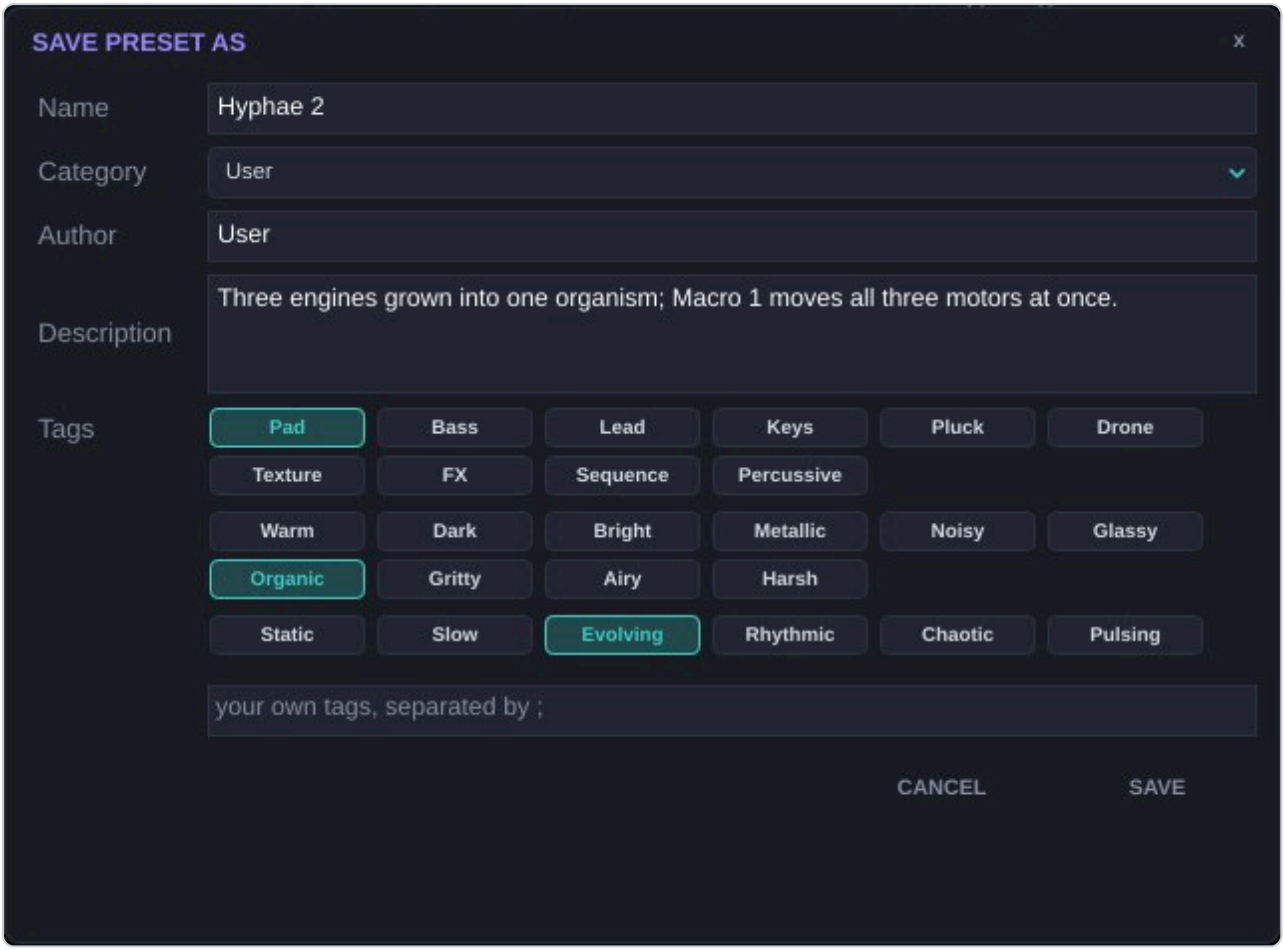
Click the preset name or **BROWSE**. The browser combines four filters:

- **Category**: *All*, *Favorites*, the 17 factory categories and those of your own presets, each with its preset count.
- **Favourites**: the heart next to each preset.
- **Tags** in three families, combined with AND (a preset must have every tag you select):
- **Type**: Pad, Bass, Lead, Keys, Pluck, Drone, Texture, FX, Sequence, Percussive.
- **Character**: Warm, Dark, Bright, Metallic, Noisy, Glassy, Organic, Gritty, Airy, Harsh.
- **Motion**: Static, Slow, Evolving, Rhythmic, Chaotic, Pulsing.
- **Search** by text in name, author, description and tags.

When you select a preset, the bottom area shows its category, author, description and tags. **LOAD** loads it.

The arrows on the top bar step through the filtered list.

16.2 Saving



SAVE AS asks for a name, category, author, description and tags, plus your own tags separated by semi-colons. **SAVE** overwrites the current user preset. Caosmo **suggests character tags** based on the engines of the active lanes (for example Glassy and Airy for Kuramoto, Metallic and Bright for Circle Map). Add Type and Motion and check them before saving. User presets are saved as `.caosmo` files in `Documents/Caosmo Presets/User`. Right-click a user preset to edit its information, rename it or delete it.

A preset stores every parameter, the routes (including those created with MIDI Learn), the four VENV shapes, the keytrack curve and the EQ snapshots. It does not store interface preferences. *Settings → OPEN PRESETS FOLDER* opens the preset folder.

16.3 The 20 categories

Category	Presets	What you'll find
Nebulae	16	Wide, evolving pads.
Fault Lines	15	Basses: double wells, saturated ladders, sub-scrolls.
Filaments	14	Leads and melodies.
Sparks	15	Plucks, mallets, percussion.
Undertow	14	Deep drones.
Static Fields	14	Noise, textures, effects.
Clockwork	14	Sequences, arpeggios and synced pulses.

Category	Presets	What you'll find
Resonant Bodies	13	Strings, bells and physical bodies.
Alloys	13	Multi-engine combinations.
Mycelia	16	Three engines behaving as one organism; Macro 1 moves all three at once.
Weather Systems	16	Attractors modulating each other: slow changes of "weather".
Bifurcations	16	A VENV drives a chaos parameter from order to turbulence; aftertouch pushes it further.
Strata	16	Low drones in which the keytrack makes high notes unstable.
Particle Fields	16	Grain clouds with Loop Lock: randomness that repeats.
Signal Decay	16	Unstable loops, frequency-shifter feedback, degradation.
Warehouse	16	Techno textures: linked LFOs, synced *gates*, stutter.
Living Machines	16	Sixteen interconnected modulators: self-organising systems.
Stochastic	12	Gendy, Noise and random triggers: Xenakis, Poisson rain, human clocks, coin-flip sequences.
Neurons & Maps	12	SuperCollider's maps played in tune and the Ecosystems models: neurons, chemical clocks, predators and prey.
Bouncing Bodies	8	Membrane: balls bouncing on drums, bars and bells.

The full list is in the appendix.

16.4 How they were made

The presets are described in text files, and a program generates, renders and measures them. Every preset is **automatically levelled** to about −18 dBFS RMS and checked so its true peak stays below −0.8 dBTP; how much the sound evolves over time is also measured, so presets tagged *Evolving* or *Chaotic* really are. That's why you can browse the bank without level jumps.

17

Tutorials

Every recipe starts from the initial patch: *Settings → RESET TO INIT*. The values are starting points: listen and adjust.

17.1 A pad that synchronises

1. L1 Engine **Kuramoto**. Ratio **Harmonic**, Spread **0.6**, N Osc **8**, Out **Coherent**.
2. ENV GEN: Attack **1.5 s**, Release **3 s**.
3. FILTER: LPF 24 dB, Cutoff **3 kHz**, Res **20 %**.
4. MODULATORS: LFO 1 on **Random Smooth**, Rate **0.08 Hz**. ASSIGN → click **K** (on MAIN). In the matrix, Amount **0.35**.
5. Raise K until you hear the sound drift in and out of lock.
6. FX1 send at **−8 dB** (bus A's Reverb).

For more life: Drift Pitch **Subtle** and, on L2, the same engine with Ratio **Golden** and Level **−9 dB**.

17.2 A double-well bass with bite

1. L1 **Duffing**: Gamma **0.28**, Delta **0.18**, Omega **1**, Out **Position**.
2. Voices **1**, Legato **On**, Glide **40 ms**.
3. FILTER: **Ladder**, Cutoff **400 Hz**, Res **35 %**, LPG **Amp + Filt**.
4. ENV GEN: Attack **2 ms**, Decay **350 ms**, Sustain **40 %**.
5. Matrix: *Velocity → L1 Motor 1* (Gamma), Amount **0.3**, Curve **Exp**. Playing hard makes the mass jump between wells.
6. OUTPUT: Drive **25 %** Tape, Bass Mono **120 Hz**.

17.3 Sand-and-metal percussion

1. L1 **Logistic/Henon**: Morph **0.3**, Smooth **0.55**.
2. ENV GEN: ADSR **One-Shot**, Attack **0.3 ms**, Decay **90 ms**, Sustain **0**.
3. FILTER: **BPF**, Res **60 %**. Matrix: *Keytrack → L1 Cutoff*, Amount **0.5**.
4. L2 **Chua** Alpha **10.5**, same envelope with Decay **40 ms**, Level **−6 dB**: the "hit".
5. Matrix: *Note Random → L1 Cutoff*, Amount **0.08**: every hit sounds different.

17.4 A cloud that rewrites itself

1. L1 **Wave Grains**: Size **60 ms**, Count **10**, Scatter **0.7**, Spray **0.9**, Jitter **0.2**.
2. Chaos **Rossler**, Chaos Rate **0.3 Hz**, Chaos>X **0.6**, Chaos>Y **0.5**, Chaos>Fold **0.3**, Tau **0.6**. Grain **Frozen**.
3. ENV GEN: Vactrol with Rise Slow **800 ms** and Fall Slow **4 s**.
4. LFO 2 on Random S&H, Sync Host **1/8**, to *L1 Motor 5* (Size ms) with Amount **0.25**. Turn on **Loop Lock** with Division **2 bars**: the grain size changes the same way every time, like a sequence.

5. In MASTER → FX, set slot 2 of bus B to **Granular** with Option **Freeze**, and raise the lane's FX2 send to **−12 dB**.

17.5 An expressive MPE cello

1. Settings: MPE **Lower**, MPE PB Range **48** (or whatever your controller uses).
2. L1 **Bowed String**: Press **0.5**, Speed **0.5**, Pos **0.18**, Body **0.6**, Rasp **0.15**.
3. ENV GEN: **Vactrol**, Rise Fast **20 ms**, Fall Slow **1.2 s**.
4. Matrix: *MPE Pressure* → *L1 Motor 2* (Speed) Amount **0.5**; *MPE Pressure* → *L1 Motor 1* (Press) Amount **0.25**; *MPE Slide* → *L1 Motor 3* (Pos) Amount **0.3**.
5. LFO 1 **Poly**, Sine **5.5 Hz**, to *L1 Fine* Amount **0.08**, with **Fade In 600 ms**: a late vibrato, like a player's.

17.6 A techno drone with the Deluge

1. L1 **Mackey-Glass**: Sharp **0.45**, Crackle **0.1**, Octave **Fixed**. L2 **Gray-Scott** with BPF **0.6**.
2. LFO 1 Sync Host **1/16**, Pulse, to *L1 Level* and *L2 Level* (negative Amount, −0.6): a rhythmic gate. Link **Dual** with Link Mode **Divide**: LFO 2 at half speed, to L2 Cutoff.
3. MASTER → DELUGE: Deluge FX **On**; Mod FX **Dimension** (Depth 20); Delay Amount **18**, Sync **1/8**, Dotted, Analog **On**; Reverb **Mutable** Amount **15**; Compressor Threshold **20**, Blend **35**.
4. In REAPER, automate **Deluge Stutter** on the last beat of every four bars.

17.7 Writing a bifurcation

1. L1 **Sprott**: Chaos **0.1**, Damp **1**, Out **X**.
2. VENV 1: draw a slow ramp from 0 to 1 with a plateau at the end. Mode **1-Shot**, Time **12 s**, Voicing **Poly**.
3. Matrix: *VENV 1* → *L1 Motor 1* (Chaos), Amount **0.7**.
4. Hold a note: the tone splits into sub-octaves and ends in turbulence. *Aftertouch* → *L1 Motor 1* with Amount **0.2** lets you push it live.

17.8 Tuned rain with Poisson triggers

1. L1 **Noise**: Type **Dust2**, Density **200**, Key BPF **0.8**, Q **18**.
2. ENV GEN: Attack **5 ms**, Release **1 s**. FILTER: **Ladder**, Cutoff **2 kHz**, Res **40 %**.
3. MODULATORS → TRIGGER 1: Mode **Dust**, Rate **6 Hz**, Output **S&H Decay**, Length **150 ms**, Dist **Beta**.
4. Matrix: *Trig 1* → *L1 Cutoff*, Amount **0.4**, and *Trig 1* → *L1 Motor 5* (Q), Amount **0.2**.
5. *Macro 1* → *Trig1 Prob*: the macro thins or thickens the rain. To give every note its own character, *Note Random* → *L1 Motor 2* with Note Random Dist on **Gauss**.

18

Performance and CPU

The cost depends on **voices × lanes × engine**. Some guidelines:

- The lightest engines are Maps, Noise, Logistic/Henon, Duffing, Sprott, Chua and Circle Map. The heaviest are Lorenz Additive (with Engine HQ), Wave Grains with a high Count, Kuramoto with 12 oscillators, and Gray-Scott.
- **Sub 1** doubles the cost of that lane's engine; **Sub 2** is free.
- **Poly** modulators cost one instance per voice.
- Lower **Voices** to what you need: 8 is usually enough for pads, 1–4 for basses and leads.
- Switch **Engine HQ** off if you don't use Lorenz Additive or are short of CPU.
- Switch unused lanes off with **PWR** (Mute does not save CPU).
- Visualisers only use CPU while the window is open; close it or pause them in Settings.
- In the DAW, render or freeze tracks in large mixes.

19

Troubleshooting

Symptom	Likely cause and fix
No sound	Does the lane have an engine and PWR on? Does the lane's MIDI channel match? Has Gray-Scott with Drift 0 died out? Bowed String with Press or Speed at 0?
Sound, but no pitch	The engine is unpitched (Lorenz+Rossler, Logistic/Henon, Gray-Scott without BPF). Raise Sub 2 or use a Comb/BPF filter with keytrack.
The note jumps an octave	Mackey-Glass on Octave Free, or Sprott/Chua in a period-doubling zone. Use Octave Fixed or lower the chaos parameter.
Clicks when playing fast	Very short Attack with an engine that starts with lots of energy. Raise Attack to 2–5 ms or use Vactrol.
Level jumps	A system crossing into chaos. Enable the master limiter or lower the chaos parameter.
Plugin missing (macOS)	Gatekeeper: run `xattr -dr com.apple.quarantine` on `Caosmo.vst3`.
Plugin missing (Windows)	`Common Files\VST3` is not in the DAW's paths, or the copy lacked permissions.
Slow with many voices	See the performance chapter.
MPE not responding	MPE Mode on, the right zone (Lower/Upper), and the DAW track receiving all channels.
A route does nothing	Check its light in the matrix: red (per-note source to a global destination) or amber (engine parameter that doesn't exist).
No stutter heard	Deluge FX **On**. Stutter repeats what has just played: switch it on while audio is sounding.

20

How it was made

Caosmo was built over five versions, from a playable core to a complete instrument with effects and binaries for three systems.

20.1 Research

The starting point was Fragua's public documentation: its three-lane architecture, the list of engines, the description of each control and the published demos. From there each system was reimplemented from its original equations in the scientific literature (Kuramoto, Duffing, Sprott, Lorenz, Rössler, May, Hénon, Chua, Gray-Scott, Mackey-Glass, Arnold), without access to Fragua's code.

20.2 Code architecture

- **Language and framework**: C++ on JUCE 8.0.4, built with CMake.
- **Parameters**: a single state tree with more than a thousand automatable parameters. The parameters of all seventeen engines always exist on every lane, which is why switching engines loses nothing.
- **Engines**: each system is integrated per sample, most with fourth-order Runge-Kutta. Pitched engines scale their integration step with the note frequency, capped to stay stable in the treble; some (Sprott, Chua) were calibrated by measuring their real period with the test bench so the note lands where it should. Mackey-Glass includes a period detector that corrects the delay in real time.
- **Numerical safety**: every engine checks its state on every sample and restarts if a variable blows up.
- **Modulation**: computed in short blocks. Per-note sources live inside each voice; global ones are shared. On every block the matrix decides which routes apply to each destination.
- **Interface**: drawn with JUCE in its own style; the modulation halos read the real modulated value from the audio thread without blocking it.

20.3 The Deluge port

The DELUGE FX effects were copied from the Deluge community firmware repository. 31 files are byte-identical to the original. To build them outside the hardware, small stand-ins were written for the firmware headers (memory, parameters, tables) along with a portable version of the Argon vector library. Only one firmware file was changed: the impulse-response processor of the analogue delay, so that computers use its scalar path, which the firmware itself documents as equivalent.

Audio is converted to the Deluge's 32-bit fixed point, processed in 64-sample blocks with the same parameter conversions as the device's menus, and converted back to floating point. Because the Deluge runs at 44.1 kHz, the effect rates are rescaled to the host sample rate so times match in milliseconds. A fixed −12.96 dB gain compensates the Deluge's output stage, so the rack at neutral does not change the level.

20.4 Presets

The 288 presets are written as text: each line sets the parameters that differ from the initial patch. A generator loads them into the real engine, renders them, measures their RMS level and true peak, levels them, checks their evolution over time and packs them into the plugin. The 128 experimental presets and the 32

from version 0.5 are produced by scripts that combines per-category design rules (which engines, which interconnected modulators, which macros).

20.5 Testing

- A dedicated test bench with modes for engine stability, parameter sweeps, CPU use, loading every preset, the interface and the Deluge rack's gain.
- AddressSanitizer and UndefinedBehaviorSanitizer builds with no errors.
- **pluginval** (Tracktion) at strictness 7, including its thread-safety test, which uncovered a Mod FX bug in version 0.4 that was then fixed.
- Continuous integration on GitHub Actions: every change is built for Linux, Windows and macOS, validated with pluginval and packaged.

21

Glossary

Term	Meaning
Attractor	The region a dynamical system with friction evolves towards.
Strange attractor	An attractor with fractal structure on which motion never repeats.
Bifurcation	A qualitative change of behaviour as a parameter varies.
Period doubling	A bifurcation where the cycle takes twice as long to repeat: the sub-octave appears.
Hysteresis	The system depends on its history: raising and lowering a control do not pass through the same states.
Arnold tongues	Parameter regions where two oscillators lock in a rational ratio.
LPG	*Low pass gate*: a gate that opens volume and brightness together, typical of Buchla.
Map	A discrete-time system: computes the next state from the current one.
MPE	*MIDI Polyphonic Expression*: each note on its own channel, with its own bend, pressure and slide.
Poly / Global	A modulator with one copy per note / a single shared copy.
Phase portrait	A plot of one variable against another: shows the shape of the attractor.
Runge-Kutta 4	A fourth-order numerical method for integrating differential equations.
Seed	A number that initialises each note's random generator.
True peak	The real peak of the reconstructed analogue signal, which can exceed the sample peaks.
Vactrol	A lamp-and-photoresistor opto-coupler, with slow response and memory.

22

Appendices

22.1 Preset catalogue

Category	Presets
Nebulae	Glass Synchrony, Cumulus Engine, Aurora Filament, Slow Bloom Vactrol, Coherent Choir, Polar Drift, Tidal Glass, Velvet Nebula, Distant Freeze, Ice Shelf, Murmuration, Halo Shift, Moss Choir, Ozone Layer, Mercury Rain, Dark Matter Pad
Fault Lines	Double Well Bass, Ladder Chaos Bass, Scroll Sub, Ember Bass, Gear Bass, Tar Pit, Rubber Bow Bass, Lorenz Growl, Reese Swarm, Formant Bass, Sub Gear Pulse, Fold Thump, Basalt, Seismic Wobble, Iron Tongue
Filaments	Thread Lead, Bifurcation Lead, Glass Whistle, Snarl Lead, Gear Shift Lead, Vocal Swarm, Ribbon Bow, Hysteresis Saw, Shard Lead, Duffing Flute, Neon Spiral, Orbit Lead, Ring Lead, Solar Wind
Sparks	Glass Pluck, Gear Mallet, Vactrol Ping, Struck String, Crackle Key, Chaos Clave, Sand Snare, Well Kick, Glass Bell, Kalimba Gears, Grain Harp, Rubber Tom, Metal Tine, Pebble Hat, Comb Pluck
Undertow	Abyssal Hum, Tanpura Swarm, Mitotic Bed, Eternal Bow, Magnetosphere, Hollow Earth, Saturn Rings, Low Pressure, Glacier, Ember Drone, Monolith, Solar Furnace, Choir of Wells, Deep Field
Static Fields	Cell Division, Hiss Map, Radio Static, Rain on Tin, Insect Swarm, Dust Storm, Broken Machine, Geiger, Whale Song, Static Choir, Wind Chimes Far, Tectonic Creak, Plasma Fizz, Electric Moss
Clockwork	Gear Train 16, Pulse Swarm, Arp Steps, Locked Random, Trance Gate Grains, Dotted Echo Pluck, Clockwork Bass, Pendulum Quad, Morse Code, Ratchet, Euclid Wells, Phase Drummer, Bouncing Ball, Swing Machine
Resonant Bodies	Cello Gesture, Piano Wire, Raucous Fiddle, Bell Foundry, Viola Section, Glass Harmonica, Duffing Drum, Rosin Burn, Kalimba Wire, Ember Strings, Monochord, Hammered Gears, Cathedral Bow
Alloys	Three Metals, Swarm and Ember, Chaos Organ, Grain Bow, Split Personality, Hybrid Reese Lead, Reaction Bells, Tectonic Plates, Spectral Strings, Machine Choir, Velocity Split, Hail and Thunder, Alloy Lead
Mycelia	Hyphae, Spore Bloom, Rhizome Choir, Fruiting Body, Lichen Clock, Root Network, Damp Cathedral, Spore Print, Mossgate, Underground Post, Truffle Engine, Decomposer, Fungal Glass, Leaf Litter, Bioluminescence, Wood Wide Web
Weather Systems	Pressure Front, Jet Stream, Isobar Drift, Cumulonimbus, Monsoon Engine, Polar Vortex, Foehn Wind, Doldrums, Katabatic, Squall Line, Dew Point, Barometric Choir, Trade Winds, Hail Front, Eye of the Storm, Thermocline
Bifurcations	Period Doubling, Feigenbaum Gate, Strange Window, Edge of Order, Logistic Cliff, Hopf Point, Saddle Node, Intermittency, Crisis Mode, Torus Breakdown, Phase Slip, Tipping Point, Unfolding, Onset of Chaos, Symmetry Breaking, Lyapunov Slope
Strata	Mantle Plume, Subduction, Basalt Hymn, Magma Chamber, Lithosphere, Continental Drift, Fault Creep, Bedrock Pulse, Abyssal Plain, Pyroclast, Moho Boundary, Seismic Hum, Graben, Deep Time, Geode, Core Sample
Particle Fields	Brownian Motion, Pollen Count, Dust Devils, Ion Trail, Photon Rain, Starling Cloud, Cloud Chamber, Snow Crystal, Plankton Bloom, Firefly Field, Static Snow, Glitter Drift, Murmur Lattice, Ash Fall, Quantum Foam, Swarm Logic

Category	Presets
Signal Decay	Tape Rot, Vinyl Ghost, Burnt Circuit, Shortwave Drift, Corroded Choir, Dead Channel, Smoulder, Rust Bloom, Lost Transmission, Bitter Static, Ember Radio, Cinder Loop, Oxidised, Faded Broadcast, Burn-In, Ghost in the Hiss
Warehouse	Klock Tower, Concrete Pulse, Strobe Memory, Sub Floor, Dark Room, Sixth Hour, Steam Pipe, Hangar Echo, Fogline, Iron Lung, Night Shift, Turbine Hall, Late Set, Machine Breath, Freight Yard, Peak Time Drone
Living Machines	Feedback Organism, Autopoiesis, Clockwork Heart, Self-Similar, Cybernetic Garden, Homeostat, Neural Tangle, Reflex Arc, Cellular Automaton, Golem Breath, Strange Loop, Emergence, Daisyworld, Metabolism, Hive Mind, Perpetual Motion
Stochastic	Xenakis Choir, GENDYN 3, Polygon Drift, S709, Dust Devils, Crackle Box, Gray Code, Pink Surf, Poisson Rain, Gauss Clock, Coin Flip Seq, Lorenz Stutter
Neurons & Maps	Latoo Bells, Standard Hum, Gingerbread Man, Cusp Organ, LinCong Steps, Henon Reed, Spiking Neuron, Belousov Clock, Brusselator Pad, Fox & Hare, ACT Turbulence, Budworm Outbreak
Bouncing Bodies	Ping Pong Drum, Marble Bar, Bell Drop, Tabla Physics, Square Skins, Low Gravity, Spring Plate, Squiz Bounce

22.2 Parameter names in the DAW

Every parameter can be automated. In the DAW they appear with a prefix showing where they live:

Prefix	Example	Block
L1, L2, L3	L2 Cutoff, L1 Kuramoto K	Lanes and their engines
LFO1–8	LFO3 Rate	LFOs
VENV1–4	VENV1 Time	Drawable envelopes
Attr1–4	Attr2 P1	Attractors
Trig1–4	Trig3 Rate	Random triggers
Slot1–24	Slot5 Amount	Matrix routes
Master / Comp / Sat / Limiter	Comp Thresh	Master chain
FX A/B/C	FX B Slot2 P1	Effect buses
Deluge	Deluge Delay Amount	DELUGE FX rack
(no prefix)	Macro 1, Voices, Glide	Globals

To automate the sound without touching presets, use the **macros**: their names are stable and every preset assigns them to what matters.

22.3 Version history

Version	Changes
0.1	Playable core: 3 lanes, 12 engines, filter, envelope, LFOs, matrix and a generic interface.
0.2	Vactrol and VENV, attractors, LFO links and Loop Lock, EQ snapshots, master chain, FX buses, tagged browser, 128 presets.
0.3	128 experimental, evolving presets.
0.4	DELUGE FX rack, GPL v3 licence, Windows and macOS, constant latency, complete manual.

Version	Changes
0.5	SuperCollider generators: Gendy, Maps, Ecosystems, Noise and Membrane engines; random triggers; distributions; LFO waves with memory; Friction and Spring filters; seven attractors; six spectral and cycle effects; 32 presets.

22.4 Licence

Caosmo is free software: you may use, study, modify and redistribute it under the **GNU General Public License version 3 or later**. This is required because it includes GPL-3.0 code from the Synthstrom Audible Deluge firmware. If you distribute binaries, you must also offer the complete source code under the same licence. JUCE 8 is used under its AGPLv3 option, which is compatible with GPLv3 through section 13 of both licences. The full licence text is in the `LICENSE` file of every package.

22.5 Credits

Part	Author and licence
Caosmo	Eloy Platas, GPL v3 or later.
Reference concept	Fragua by Pura Belia (purabelia.com). Caosmo is an independent work.
DELUGE FX	Synthstrom Audible Deluge community firmware, © Synthstrom Audible Limited and contributors, GPL v3.
Mutable reverb	Based on code by Emilie Gillet (Mutable Instruments), MIT licence.
Freeverb	Jezar at Dreampoint, public domain.
Framework	JUCE 8.0.4, AGPLv3.
Validation	pluginval by Tracktion.
0.5 generators	Reimplemented from SuperCollider (GPL v2 or later): Gendy1 (Nick Collins, after Iannis Xenakis), the core ChaosGen and PV_ UGens, SLUGens (Nick Collins), MCLDUGens (Dan Stowell: Friction, TBall, WaveLoss, Squiz), ChaosUGens (Till Bovermann) and BhobUGens (GaussTrig, LorenzTrig).
Dynamical systems	Kuramoto (1975), Duffing (1918), Sprott (1997), Lorenz (1963), Rössler (1976), May (1976), Hénon (1976), Chua (1983), Gray and Scott (1983), Mackey and Glass (1977), Arnold (1961), Xenakis (1991), Pickover (1994), Chirikov (1979), Devaney (1984), FitzHugh (1961) and Nagumo (1962), Prigogine and Lefever (1968), Field and Noyes (1974), Lotka (1925) and Volterra (1926), Arneodo, Coullet and Tresser (1981), Ludwig, Jones and Holling (1978).